AN EMBEDDED BAND FOLIATED BY CLOSED ASYMPTOTIC LINES AND NONRIGID SMOOTH TIGHT TORI

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Abstract. We construct a smooth embedded negatively curved band foliated by closed asymptotic curves. The full return map of its nonruling asymptotic family is the identity, and its compactly supported infinitesimal bendings are parametrized by arbitrary smooth functions of a transverse leaf coordinate. We complete the band to a smooth embedded tight torus while retaining its bending region. Opposite signs of one retained bending give two noncongruent tight embeddings with exactly the same intrinsic metric; they agree on a nonempty open set and have no open planar subset. Countably many disjoint bendings give a Cantor family with these properties.

The construction separates spatial closure from cancellation of the asymptotic return period. A corrugated spherical normal loop and a positive speed provide an embedded identity-return band with two visible exits. Visible connectors, differential Legendre duality, and radial saddle filling then produce explicit end profiles. Compactification and convex attachment close the surface, and an affine marking removes its ambient symmetries. The same band also gives fixed-boundary isometric saddle annuli, an annulus with injective Gauss map and zero normal linking along its closed asymptotic curves, and a complete properly embedded negatively curved cylinder with total absolute curvature 4π. These constructions connect the smooth tight-congruence question with the geometry and return dynamics of closed asymptotic curves.

1. Introduction

1.1. **Congruence, tightness, and the historical problem.** When does the intrinsic metric of a surface determine its image in $\mathbb{R}^3$ up to a Euclidean motion? Convex-surface rigidity provides the classical model, but a surface with both positive and negative curvature poses a different problem. The negative-curvature region supports a hyperbolic system, and its asymptotic curves are the characteristics. Yau emphasizes this connection between metric uniqueness and asymptotic curves in his review of the global embedding problem [12, pp. 235–236].

Tight surfaces form a natural class in which to test uniqueness beyond convexity. For a closed embedded surface, tightness is equivalent to the two-piece property (connected intersection with every open half-space) and to attaining the topology-dependent lower bound for total absolute curvature. A tight torus has total absolute curvature 8π. Aleksandrov proved congruence for analytic tight tori in 1938. Nirenberg's 1963 work obtained a smooth partial result. Its torus formulation assumes C^5 regularity and $\nabla K \neq 0$ at $K = 0$, together with an additional asymptotic-curve hypothesis. The two torus versions are recorded together in [10, Theorem 1.6.1]; the original smooth argument is [11].

The distinction between smooth and polyhedral surfaces is already present in the earlier literature. Banchoff constructed noncongruent isometric tight polyhedral tori; Banchoff–Kühnel record these examples and then ask whether two isometric smooth tight tori must be congruent [10, Example 1.6.2

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and Question 11]. Ghomi's later survey states the smooth tight-rigidity problem and its fixed-boundary negative-curvature annulus counterpart [1, Problems 1.2–1.3]. Thus the smooth tight-congruence question has a specific history; nonuniqueness for arbitrary smooth surfaces is an older phenomenon.

Han–Khuri subsequently proved two rigidity theorems that allow several closed asymptotic curves. Besides their hypotheses at the boundary of the negative-curvature region, both theorems require

$$\oint_\Gamma k_g k_n |K|^{-1/2}\, ds \neq 0$$

on every closed asymptotic curve; k_n is the normal curvature in the tangent direction perpendicular to Γ. They also ask whether tightness itself forces this nonvanishing [14, Theorems 1–2 and the following remark]. Our protected band has identity return, so this integral vanishes on every one of its leaves. The resulting example therefore lies outside the nondegeneracy hypothesis of those theorems.

Our main result gives two smooth embedded tight tori with one common metric, noncongruent images, and an open region of pointwise agreement. The entire change is supported inside a compact negative-curvature region. We also obtain a Cantor family of such realizations. Here two embedded surfaces are *congruent* if an ambient Euclidean isometry carries one image onto the other. The following statement therefore rules out congruence of images, as well as congruence of parametrized maps. Write g_{Euc} for the Euclidean metric.

Theorem 1.1 (Localized nonuniqueness for tight tori)**.** *There are a smooth metric g on T^2 and two smooth tight embeddings $X_+, X_- : T^2 \to \mathbb{R}^3$ such that*

$$X_+^* g_{\mathrm{Euc}} = X_-^* g_{\mathrm{Euc}} = g.$$

Their images are noncongruent, they agree on a nonempty open subset, and neither has an open planar subset.

In fact, the same metric can be chosen to admit a continuous family

$$\varepsilon \longmapsto X_\varepsilon \in \mathrm{Emb}_g(T^2, \mathbb{R}^3), \qquad \varepsilon \in \{-1,1\}^{\mathbb{N}},$$

of tight embeddings with pairwise noncongruent images. All members agree on the same nonempty open set, have no open planar subset, and have total absolute curvature 8π. The induced map into $\mathrm{Emb}_g(T^2,\mathbb{R}^3)/\operatorname{Isom}(\mathbb{R}^3)$ is a topological embedding.

Here $\mathrm{Emb}_g(M,\mathbb{R}^3)$ is the space of smooth embeddings inducing g, with the C^∞ topology; $\operatorname{Isom}(\mathbb{R}^3)$ acts by postcomposition. The sign space has its product topology and is homeomorphic to the Cantor set. The open set in the theorem is fixed by the construction. The family is continuous on a Cantor parameter space, which is totally disconnected. It supplies no nonconstant path of isometric embeddings. In particular, the theorem does not settle the continuous smooth-flex problem stated in [1, Problem 1.1].

1.2. **Why localized bendings give exact pairs.** The exact pair comes from a quadratic identity. For an immersion X and a vector field $y : M \to \mathbb{R}^3$, the pullback tensor satisfies the exact identity

$$g_{X+y} = g_X + S_X y + \langle dy, dy\rangle, \tag{1.1}$$

where $S_X y$ is the linearized metric change. If $S_X y = 0$, then $g_{X+y} = g_{X-y}$. Thus a nonzero infinitesimal bending gives an exact sign ambiguity, provided both perturbed maps remain embeddings. Their common metric is generally different from g_X.

If finitely many such fields have disjoint supports, then

$$g_{X+\sum_{j=1}^m \varepsilon_j a_j y_j} = g_X + \sum_{j=1}^m a_j^2 \langle dy_j, dy_j\rangle, \qquad \varepsilon_j \in \{-1,1\}.$$

This is the finite sign mechanism. The Cantor family follows by choosing countably many disjoint fields and rapidly decreasing amplitudes. The central issue is therefore geometric: construct a tight torus carrying localized bendings in a compact negatively curved region, and eliminate ambient symmetries on the region that stays fixed.

1.3. **The band that carries the construction.** Our bending region lies in a principal-normal ruled band

$$X(s,u) = \gamma(s) + uE(s).$$

Here s is arclength along a closed curve, and the periodic frame (T, E, n) satisfies $T_s = kE$, $E_s = -kT + \tau n$, and $n_s = -\tau E$, with $\tau \neq 0$. The band has strictly negative Gaussian curvature. Put $\rho = \sqrt{|\tau|}$ and $D = k_s - k\tau_s/\tau$. Its nonruling asymptotic curves have return map

$$\mathcal{R}(u) = \frac{u}{1 + \rho(0) I_0 u}, \qquad I_0 = \oint \frac{D}{2\rho}\, ds. \tag{1.2}$$

The multiplier $\mathcal{R}'(0)$ is always one. The stronger condition $I_0 = 0$ makes the full return map the identity, so nearby characteristics are closed and admit a single-valued transverse parameter. Theorem 3.3 proves that this condition is exactly what permits compactly supported infinitesimal bendings on a thin one-sided band. In the identity case, every such bending comes from one arbitrary transverse test function.

Producing that band and completing it are separate tasks. For a prescribed spherical normal loop, spatial closure is a vector moment condition, whereas identity return is one scalar period condition. Theorem 4.2 separates them and cancels the period without losing closure. The explicit corrugated seed supplies additional embedding and two-sided visibility properties. We then protect a compact subannulus, alter the support potential only outside it, and attach the remaining geometry.

Figure 1 shows the thin balanced band in the saved numerical atlas and a local view of its protected region. On the unchanged region, its rescaled leaf label c is related to the first integral used below by $v = (\sigma c)^{-1}$, with $\sigma = 10^{-6}$ in this numerical model. The unrolled strip resolves leaves that cannot be separated in the full physical view. The explicit analytic seeds used in the proof are drawn separately in Figure 4.

1.4. **Further consequences and earlier questions.** The same protected band has two useful completions besides the tight torus.

Theorem 1.2 (Closed asymptotic annulus)**.** *There is a smooth embedded compact annulus $N \subset \mathbb{R}^3$ with $K < 0$ in its interior. Each boundary component is a circle whose plane is tangent to N along that circle. For a suitable normal, its Gauss map is a diffeomorphism*

$$N^\circ \longrightarrow \mathbb{S}^2 \setminus \{\pm e_3\}.$$

An open subannulus is foliated by embedded closed asymptotic curves. Each has linking number zero with a sufficiently small normal push-off.

Theorem 1.3 (Complete embedded saddle cylinder)**.** *There is a smooth proper embedding $X_\infty : S^1 \times \mathbb{R} \to \mathbb{R}^3$ whose image Σ has complete induced metric and $K < 0$ everywhere. It contains an open annulus of embedded closed asymptotic curves with identity return and zero normal linking. A choice of normal gives*

$$n : \Sigma \longrightarrow \mathbb{S}^2 \setminus \{\pm e_3\} \quad \text{a diffeomorphism,} \qquad \int_\Sigma |K|\, dA = 4\pi.$$

For some nonempty open interval I, there is an injective linear family of infinitesimal bendings parametrized by $C_c^\infty(I)$, all supported in one fixed compact subset of Σ.

The relation to the published questions is as follows.

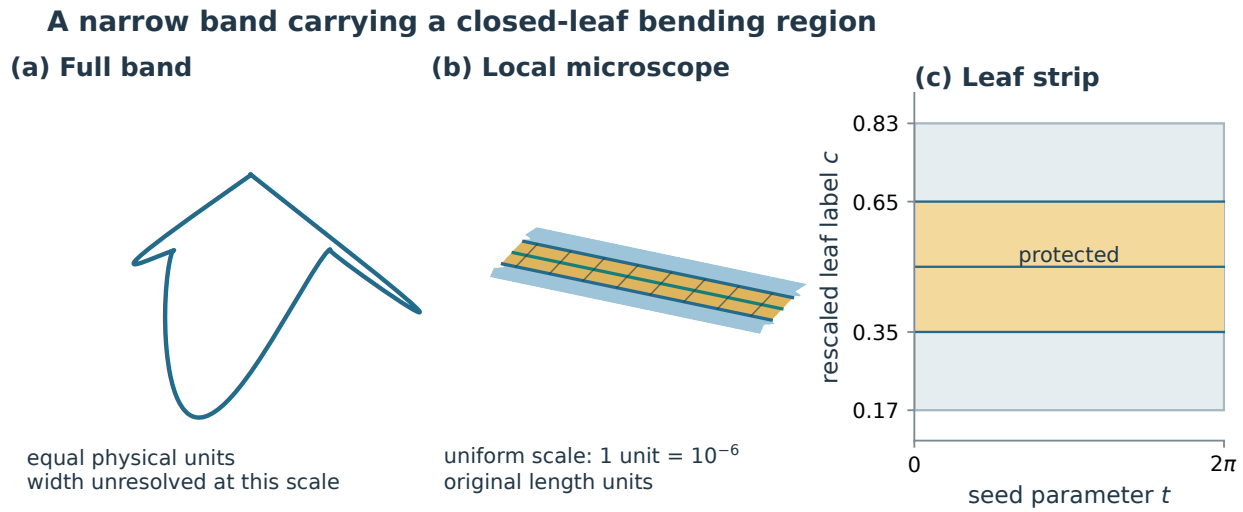


FIGURE 1. The balanced numerical band used by the atlas's saved completion. (a) The complete band, with a protected closed leaf in blue. Its actual width is too small to resolve at this scale. (b) A rigidly rotated, uniformly magnified local patch: one displayed length unit equals 10^{-6} original units. Gold marks the unchanged closed-leaf region; blue and teal curves are its asymptotic leaves, and the short transverse lines are straight rulings there. (c) An unrolled parameter diagram, rather than a physical projection, with protected levels $0.35 \le c \le 0.65$ and the two ends identified. This saved balanced numerical example illustrates the construction; the analytic existence argument is Lemma 5.2 and Theorem 5.5.

(i) Da Cruz–Garcia record the question of a cylindrical region foliated by closed asymptotic lines [4, p. 17]. The identity-return band provides such a region in a surface in $\mathbb{R}^3$. Their paper studies asymptotic lines of general plane fields; our construction is on an actual embedded surface.

(ii) Nirenberg's annulus question concerns closed asymptotic curves on a negatively curved annulus with convex planar boundaries, as explained in [1, §1.2]. Theorem 1.2 provides an open band of closed leaves in an annulus of this type.

(iii) Ghomi–Raffaelli ask whether an embedded negatively curved surface with injective Gauss map can have a closed asymptotic curve with zero normal linking [13, Problem 1.1]. Our annulus has both properties. The surface normal along a leaf Γ is a continuous binormal in the sense of [1, §1.2]: it is orthogonal to both Γ' and Γ'', including at inflections. Its injectivity and zero linking give the example asked about in [1, Problem 1.4]. Ding's recent open-annulus construction also proves the zero-linking and regular-binormal conclusions [8, Theorem 1.1 and Corollary 1.2]. The boundary geometry and closed-leaf foliation in Theorem 1.2 give additional conclusions.

(iv) Theorem 1.1 gives a negative answer to the smooth tight-torus congruence question [10, Question 11]. Corollary 10.3 gives noncongruent isometric negative-curvature annuli that agree on full boundary collars, addressing [1, Problem 1.3].

(v) The continuum of closed leaves shows that the closed-curve restriction discussed by Yau [12, p. 236] does not hold for all smooth tight tori. Corollary 10.2 gives the vanishing integral relevant to the remark following Han–Khuri's rigidity theorems [14, p. 464].

(vi) Ni–Zhang–Zhang construct a complete negatively curved immersed cylinder containing a closed asymptotic curve and explicitly leave embedded existence open [9, Theorem 3.4 and the following paragraph]. Theorem 1.3 provides a proper embedded example.

These are questions and formulations in the cited works. The geometry needed to meet their hypotheses is proved below: embedding and global Gauss-map injectivity, full boundary collars, zero linking, and completeness are separate conclusions of the construction.

1.5. **Relation to earlier constructions.** Closed asymptotic curves and principal-normal ruled strips are established parts of surface theory. Kovaleva's 1968 construction gives a negatively curved tube-type surface containing a closed asymptotic line [2]; its role in the principal-normal strip construction is recalled in [13, Note 2.1]. Garcia–Sotomayor study the qualitative behavior and structural stability of periodic asymptotic lines [3]. Our use of principal-normal rulings builds on this earlier setting. The additional local requirement is identity of the full return germ, which gives a whole band of closed leaves and allows us to classify its compactly supported bending fields.

The topological restrictions are equally important. Ghomi–Raffaelli relate normal linking to the writhe and rotation of projections of closed asymptotic curves. They discuss earlier examples with zero linking and noninjective Gauss map, and construct examples with injective Gauss map and nonzero linking [13, §§1,4]. Their Problem 1.1 asks whether zero linking and Gauss-map injectivity can occur together. Their analysis also explains why injective normals force inflections [13, Note 5.1]. This is why the signed frame in our ruled band allows $k = 0$: requiring a classical Frenet frame everywhere would exclude the geometry under consideration.

Smooth isometric nonuniqueness also predates the present construction. Sa Earp–Toubiana give closed smooth surfaces of arbitrary genus with nondiscrete isometric realizations, including families parametrized by nowhere dense compact sets [5, Theorem 2]. Their construction reflects pieces across planar seams at which the profile is flat to every order. A Cantor set of realizations is therefore not by itself a new phenomenon. Nor does a planar seam imply an open planar patch; the distinction between these conditions matters when comparing examples.

Sabitov constructs two noncongruent isometric negatively curved surfaces with a common open set [6]. His later work describes metrics of revolution with a discontinuum of noncongruent isometric immersions [7]. Local agreement and large sets of isometric realizations thus have their own earlier examples. The contribution here is their realization inside smooth embedded *tight tori*, with the changes supported in strict negative curvature, together with a band whose full asymptotic return and supported bending space are explicitly controlled. The absence of open planar patches is one of the proved geometric properties; it is not used as a blanket distinction from all earlier constructions.

Ding's October 2026 preprint constructs an embedded open negatively curved annulus with injective Gauss map and a closed asymptotic curve of zero normal self-linking [8, Theorem 1.1]. It also gives a regular injective continuous binormal [8, Corollary 1.2]. Its planar curves are obtained by simultaneous polygonal rounding and mixed-integral cancellation; the theorem makes no completeness claim and prescribes no tangent-plane boundary geometry. The analytic seed used here supports a different continuation: we cancel the full return period, classify the supported kernel, and retain a closed-leaf band in the compact annulus, tight torus and complete cylinder. The overlapping open-annulus and binormal conclusions are therefore also stated in this recent preprint, alongside the older results described above.

1.6. **Proof architecture.** The construction has two layers. First we manufacture the localized bending region. Then we complete and use it. The dependencies are:

$$\begin{gathered}\text{normal loop and closing speed} \longrightarrow \text{period balance} \longrightarrow \text{identity-return ruled band},\\ \text{corrugated seed} \longrightarrow \text{embedding and two-sided visibility},\\ \text{protected band} \longrightarrow \text{positive exits} \longrightarrow \text{circular gradient seams},\\ \text{Legendre duality and radial filling} \longrightarrow \text{parabolic end collars},\\ \text{saddle annulus} \longrightarrow \text{asymmetric tight closure} \longrightarrow \text{exact sign pairs and families}.\end{gathered}$$

Section 3 proves the local kernel theorem; Section 4 handles closure and period balance. Section 5 uses the seed to build the embedded core, and Section 6 creates its positive exits. Sections 7 and 8 complete the support and assemble the tight torus. Section 9 first proves finite sign branching, then its countable extension. Section 10 proves the stated consequences. The general projective ruled theory and the quantitative seed estimates are in Appendices A and B; neither interrupts the main construction.

2. Conventions and geometric tools

2.1. **Metrics, bendings, and tightness.** Maps between finite-dimensional manifolds are smooth unless stated otherwise; on a manifold with boundary, smoothness means local smooth extension across the boundary. A closed surface is compact, connected, and without boundary. Surface metrics are Riemannian. A compact annulus includes its boundary; A° denotes its interior. The support of a function or field is the closure in its source of its nonzero set. For an open interval or surface, C_c^∞ denotes smooth functions with compact support. Profiles in $C_c^\infty(I)$, for an open interval $I \subset \mathbb{R}$, are extended smoothly by zero to $\mathbb{R}$. For maps $X, X' : M \to \mathbb{R}^3$, a perturbation X' is supported in V when $\operatorname{supp}(X' - X) \subset V$. Writing $V \Subset U$ means that the closure of V is compact and contained in U. The torus T^2 is $S^1 \times S^1$, with $S^1 = \mathbb{R}/2\pi\mathbb{Z}$. Norms on compact sources use fixed smooth atlases or a fixed background metric. For any smooth map $X : M \to \mathbb{R}^3$, write $g_X = X^* g_{\mathrm{Euc}}$; this is a Riemannian metric when X is an immersion. For an oriented immersion with unit normal N, we use

$$g_X(v, w) = \langle dX(v), dX(w) \rangle, \qquad \mathrm{II}_X(v, w) = \langle D_v(dX(w)), N \rangle.$$

Here D is the Euclidean connection, applied to the ambient vector field $dX(w)$; the displayed normal component is independent of the extensions of v, w. The shape operator is defined by $g_X(A_X v, w) = \mathrm{II}_X(v, w)$, so $dN(v) = -dX(A_X v)$ and $K_X = \det A_X$. The Gauss map is the chosen unit normal $N : M \to \mathbb{S}^2$. Changing the normal changes the sign of II_X and A_X, and leaves K_X and the asymptotic directions unchanged. We write dA_X for the area measure of g_X, abbreviating it to dA when the immersion is fixed. An open planar subset is an image $X(U)$ contained in an affine plane, where $U \subset M$ is nonempty and open. The strain is the symmetric tensor

$$S_X y(v, w) = \langle dX(v), dy(w) \rangle + \langle dX(w), dy(v) \rangle.$$

An *infinitesimal bending* is a field $y : M \to \mathbb{R}^3$ with $S_X y = 0$; this definition allows both tangential and normal components. We write $\mathcal{G}(X) = g_X$ for the metric reconstruction map. The tensor $\langle dy, dy \rangle$ means $(v, w) \mapsto \langle dy(v), dy(w) \rangle$.

An asymptotic direction is a nonzero vector v with $\mathrm{II}_X(v, v) = 0$. When $K < 0$, these directions form two transverse smooth line fields. A regular curve is asymptotic when its tangent is asymptotic everywhere.

Definition 2.1 (Return and identity return)**.** Let Γ be an oriented embedded closed leaf of a smooth line field on an annulus. Choose collar coordinates $(s, t) \in (\mathbb{R}/L\mathbb{Z}) \times (-\epsilon, \epsilon)$, $L > 0$, with $\Gamma = \{t = 0\}$ and s increasing in the chosen orientation. After shrinking the collar, the field is spanned by $\partial_s + f(s, t)\partial_t$, where $f(s, 0) = 0$. For all sufficiently small t_0, solve

$$t_s = f(s, t), \qquad t(0; t_0) = t_0, \qquad \mathcal{R}(t_0) = t(L; t_0).$$

This local diffeomorphism of the transversal $s = 0$ is the *return map*. Its *return germ* identifies maps agreeing on a neighborhood of 0. The leaf has *identity return* if $\mathcal{R}(t_0) = t_0$ on some open interval about 0. Its *return multiplier* is $\mu = \mathcal{R}'(0) > 0$. Changing the transversal or its coordinate conjugates the return germ, so identity return and the multiplier are independent of those choices. Return is *attracting* if its forward iterates are defined and converge to 0 for every sufficiently small t; it is *repelling* if its inverse is attracting. For the chosen circuit direction, $0 < \mu < 1$ and $\mu > 1$, respectively, imply these properties.

A closed connected embedded surface is *tight* if its intersection with each open half-space is connected whenever nonempty. Its total absolute curvature satisfies [10, §1.1]

$$\int_M |K|\, dA \geq 2\pi(4-\chi(M)), \qquad \text{with equality exactly for tight embeddings.} \tag{2.1}$$

For a torus the equality value is 8π.

We repeatedly use openness of compact embeddings in the C^1 topology. Local projections to tangent planes exclude collisions between nearby source points, uniformly on a finite cover; compactness separates the remaining pairs. The same argument shows that an immersion injective on a compact embedded curve is injective on a sufficiently thin neighborhood of that curve.

For an oriented embedded closed curve Γ and a smooth nowhere-zero field ν transverse to its tangent, orient the small push-off by the same parameter and define

$$\mathrm{Lk}(\Gamma, \nu) = \mathrm{Lk}(\Gamma, \Gamma + \epsilon\nu)$$

for sufficiently small $\epsilon > 0$, when the two curves are disjoint. The integer is independent of this choice. We compute linking by intersection with an oriented spanning surface [16]. For the surface unit normal $\nu = N$, this is the *normal self-linking* of Γ.

2.2. **Support coordinates.** The completion is easiest to control by using the normal as the source coordinate. Let $\Omega \subset \mathbb{S}^2$ be a connected open domain with round metric g_0, and let $H : \Omega \to \mathbb{R}$. At $q \in \Omega$, identify $T_q\mathbb{S}^2$ with $q^\perp \subset \mathbb{R}^3$ and define

$$a_H = \nabla H + Hq, \qquad B_H = \nabla^2 H + H\mathrm{Id}, \qquad R_H = \det B_H. \tag{2.2}$$

The gradient and Hessian in this formula use the round metric g_0. In the sum with $H\mathrm{Id}$, $\nabla^2 H$ denotes the endomorphism obtained by raising one index of the covariant Hessian. Equivalently, the associated bilinear form is $\mathrm{Hess}_{g_0} H + H g_0$. We write $B_H(v, w) = g_0(B_H v, w)$ when a bilinear evaluation is needed. The determinant R_H is computed in a g_0-orthonormal frame. The function H is the *support potential* of this reconstruction: $H(q) = a_H(q) \cdot q$.

Proposition 2.2 (Support reconstruction)**.** *The $\mathbb{R}^3$-valued differential of a_H is B_H. Where B_H is invertible, a_H is an immersion with normal q, metric $g_{a_H}(v, w) = g_0(B_H v, B_H w)$, second fundamental form $\mathrm{II}_{a_H}(v, w) = -g_0(B_H v, w)$, and curvature R_H^{-1}. The shorthand B_H^2 for its metric refers to this bilinear form. Every oriented immersion whose Gauss map is injective with invertible differential has this representation on its Gauss image.*

Proof. Differentiating ∇H in $\mathbb{R}^3$ produces the normal term $-dH\, q$, which cancels the corresponding term from $d(Hq)$. The remaining differential is $\nabla^2 H + H\mathrm{Id}$. It is tangent to $q^\perp$, so q is normal. Differentiating $da_H(w) \cdot q = 0$ gives $\mathrm{II}(v, w) = -\langle B_H w, v\rangle$. Thus the shape operator is $-B_H^{-1}$ and its determinant is R_H^{-1}. Conversely, in Gauss coordinates set $H = X \cdot q$. Then $dH(v) = X \cdot v$, because $dX(v) \cdot q = 0$, and the tangential and normal parts of X are ∇H and Hq. □

A field B of tangent endomorphisms, self-adjoint with respect to g_0, is *Codazzi* when $(\nabla_v B)w = (\nabla_w B)v$ for all tangent vectors v, w.

Proposition 2.3 (Codazzi realization and periods)**.** *For a smooth self-adjoint field B on Ω, the vector-valued form $\beta_B(v) = Bv$ is closed exactly when B is Codazzi. On a simply connected domain this is equivalent to $B = B_H$, with H unique modulo functions $q \mapsto a \cdot q$, $a \in \mathbb{R}^3$. On a general domain one must also require*

$$\oint_\gamma \beta_B = 0 \quad \textit{for every closed curve } \gamma.$$

Proof. The normal part of $d\beta_B(v, w)$ is $-\langle v, Bw\rangle + \langle w, Bv\rangle = 0$ by self-adjointness. Its tangential part is $(\nabla_v B)w - (\nabla_w B)v$, the Codazzi defect. If the form has a primitive a, put $H = a \cdot q$. Since da is tangent, $dH(v) = a \cdot v$, hence $a = \nabla H + Hq$ and $B = B_H$. The primitive is unique up

to a constant vector, which changes H by a linear function. Vanishing periods are precisely the obstruction to a global primitive. □

On the northern hemisphere $\{q \in \mathbb{S}^2 : q_3 > 0\}$ introduce *gnomonic coordinates*

$$q(z) = \frac{(z,1)}{w(z)}, \qquad w(z) = \sqrt{1+|z|^2}, \qquad G(z) = w(z)H(q(z)).$$

In these planar source coordinates the immersion is

$$X_G(z) = (\nabla G(z), G(z) - z \cdot \nabla G(z)), \qquad K_{X_G} = \frac{1}{w^4 \det D^2 G}. \tag{2.3}$$

Indeed, writing $A = D^2 G$ gives

$$dX_G(\xi) = (A\xi, -z \cdot A\xi), \qquad g_{X_G} = A^T(I + zz^T)A, \qquad \mathrm{II} = -A/w.$$

Thus $R_H(q(z)) = w^4 \det D^2 G$. A *saddle support* is a support potential with $R_H < 0$, equivalently $\det D^2 G < 0$ in planar coordinates. Planar gradients below are Euclidean, in contrast with the spherical gradient in (2.2).

If p is a regular closed curve in the domain of G, let $\gamma = \nabla G \circ p$ be its gradient trace and $g = G \circ p$ its value trace. Then

$$g' = \gamma \cdot p', \qquad \oint \gamma \cdot dp = 0, \tag{2.4}$$

and, along the spherical curve $\widehat{q}(s) = q(p(s))$,

$$B_H(\widehat{q}', \widehat{q}') = \frac{p' \cdot \gamma'}{w(p)}. \tag{2.5}$$

We call the paired source and gradient trace (p, γ) *positive* when $p'(s) \cdot \gamma'(s) > 0$ at every parameter value. Equivalently, a spherical support-domain curve q is positive when $B_H(q', q') > 0$ everywhere. For curves in the northern hemisphere, orientation, interior, and exterior refer to their gnomonic images. A regular seam is *noncharacteristic* when its tangent is nowhere asymptotic; the displayed identity makes every positive trace noncharacteristic.

A collar is an open tubular neighborhood of a specified curve (or a one-sided neighborhood at an actual boundary). Two functions or maps have the same *germ along a seam* if they agree on a neighborhood of the entire seam in their common source. Retaining that germ means preserving this agreement on some such neighborhood. For a radial end, an *end germ* means agreement for all sufficiently small $r > 0$, all sufficiently small $r - a > 0$, or all sufficiently large r, as appropriate.

Matching *first jets* along a seam means equality of values and differentials there. A region is *protected* when subsequent modifications agree with the specified reference map or potential on an open neighborhood of that region, using the fixed source identification and the stated ambient normalization.

2.3. **Global injectivity, duality, and affine transport.** Local negative determinant will give local inversion. The following degree argument is what upgrades it to global injectivity.

Lemma 2.4 (Annular degree criterion)**.** *Orient a compact annulus and its boundary by the induced boundary orientation. Let F be a smooth map from this annulus to $\mathbb{R}^2$. Suppose the interior Jacobian has a constant nonzero sign, the boundary maps are homeomorphisms onto disjoint nested Jordan curves, and their induced winding numbers about a point inside the inner curve are $+1$ and -1. Then F is a diffeomorphism of interiors onto the intervening planar annulus and a homeomorphism of closures. The boundary Jacobian may vanish.*

Proof. For a target off the boundary images, all preimages are interior and the degree is defined by compactness. The boundary winding sum has absolute value one between the curves and is zero in the complementary regions. All local degrees have the same sign, so there is exactly one preimage

between the curves and none outside. No interior point maps to a boundary image: local openness there would produce a target on the forbidden side. Together with the boundary hypotheses this gives a continuous bijection of compact closures, hence a homeomorphism. Local inversion gives the smooth interior inverse. □

If $\nabla G : U \to V$ is a diffeomorphism between planar open domains, its *differential Legendre transform* $G^* : V \to \mathbb{R}$ is

$$G^*(y) = z \cdot y - G(z), \qquad y = \nabla G(z).$$

If $z = (\nabla G)^{-1}(y)$, differentiation gives $\nabla G^*(y) = z$ and $D^2G^*(y) = (D^2G(z))^{-1}$. In particular, $\det D^2G^*(y) = 1/\det D^2G(z) < 0$ on a saddle domain. The transform is defined on the gradient image, and its inverse transform recovers G with the chosen additive constant. This use of Legendre duality requires a global gradient inverse on the domain in question; negative determinant alone supplies only a local inverse.

We fix the planar maps and unit vectors

$$J(x_1, x_2) = (-x_2, x_1), \qquad S(x_1, x_2) = (x_1, -x_2), \qquad A_* = JS, \qquad e_\theta = (\cos\theta, \sin\theta).$$

Planar determinants use the standard orientation, and positively oriented Jordan curves run counterclockwise around their bounded complementary component. Thus $e'_\theta = Je_\theta$ and $SJ = -JS$. For an invertible affine map $x \mapsto Ax + b$, the correct transport of a bending is

$$\widetilde{X} = AX + b, \qquad \widetilde{y} = A^{-T}y. \tag{2.6}$$

Derivative pairings are unchanged, so $S_{\widetilde{X}}\widetilde{y} = S_X y$. With $\widetilde{N} = A^{-T}N/|A^{-T}N|$, the second fundamental form is $\mathrm{II}_{\widetilde{X}} = \mathrm{II}_X/|A^{-T}N|$. Its scalar factor is positive. An affine change therefore preserves curvature signs and asymptotic directions, while generally changing the metric and the curvature values. The formula for $\widetilde{y}$ preserves the strain equation; it does not claim that AX and X have the same metric. Affine maps also preserve tightness, since inverse images of half-spaces are half-spaces. See [15] for the broader projective rigidity correspondence.

3. Identity return and localized bendings

This section isolates the local mechanism. We use *asymptotic holonomy* for the return germ in Definition 2.1. Identity return is compatible with strictly negative curvature. It supplies a transverse leaf parameter, and the ruled equations turn that parameter into the full supported bending space.

3.1. **A transverse coordinate for closed leaves.**

A *first integral* of a line field is a smooth function constant on its leaves; it is *regular* when its differential never vanishes. A defining one-form α is nowhere zero and has that line field as its kernel. An integrating factor is a smooth nowhere-zero function m such that $m\alpha = dv$ for a first integral v.

Proposition 3.1 (Product and first-integral criterion)**.** *For a smooth nonsingular line foliation near an embedded closed leaf in an annulus, the following conditions are equivalent after shrinking the neighborhood U: the return germ is the identity; there is a diffeomorphism $S^1 \times J \to U$ taking each $S^1 \times \{c\}$ to a leaf, for an open interval J; there is a regular first integral $v : U \to J$ whose fibers are connected leaves. Equivalently, a defining one-form has an integrating factor.*

Proof. Start with a coordinate on a short transversal and extend it by transport along leaves. It is single-valued after one circuit exactly when return is the identity. It then defines the submersion and the local product. A product has identity return. The differential of the submersion is a nonzero multiple of any defining form. Conversely, an exact nonvanishing defining form is the differential of a transverse coordinate; near the original compact leaf, its level sets are circles. □

The proposition concerns only the foliation. A general saddle surface need not have a supported bending for every transverse function. The ruled model below provides that additional fact.

3.2. **The principal-normal model and its return.** Let $\gamma : \mathbb{R}/L\mathbb{Z} \to \mathbb{R}^3$ be a unit-speed closed curve, with $L > 0$. Assume its periodic positively oriented orthonormal frame (T, E, n), with $T = \gamma_s$, satisfies

$$T_s = kE, \qquad E_s = -kT + \tau n, \qquad n_s = -\tau E, \qquad \tau \neq 0. \tag{3.1}$$

The curvature k is signed and may vanish. Here τ is the smooth frame coefficient in (3.1). It agrees with Frenet torsion where $k \neq 0$; the frame equations define it through the inflection points. Set

$$X(s,u) = \gamma(s) + uE(s), \quad \rho = \sqrt{|\tau|}, \quad \lambda = \tau_s/\tau, \quad D = k_s - k\lambda.$$

The source band is $(\mathbb{R}/L\mathbb{Z}) \times (-b, b)$, with $b > 0$; the one-sided band below is $(\mathbb{R}/L\mathbb{Z}) \times (0, b)$. The asymptotic foliations and bending supports are taken on this source. For the normal determined by $X_s \times X_u$, differentiation gives

$$\begin{aligned} g_X &= \mathcal{E}\, ds^2 + du^2, & \mathcal{E} &= (1-ku)^2 + \tau^2 u^2, \\ \mathrm{II}_{ss} &= \frac{u\tau(\lambda + uD)}{\sqrt{\mathcal{E}}}, & \mathrm{II}_{su} &= \frac{\tau}{\sqrt{\mathcal{E}}}, & \mathrm{II}_{uu} &= 0. \end{aligned} \tag{3.2}$$

In particular $K = -\tau^2/\mathcal{E}^2 < 0$. One asymptotic family consists of the rulings. The other is spanned by

$$C = \partial_s - \frac{u}{2}(\lambda + uD)\partial_u.$$

A *complete leaf in the one-sided band* is a characteristic graph defined over the entire parameter circle and contained in $0 < u < b$. Along its characteristic graphs with $u \neq 0$,

$$u_s = -\frac{u}{2}(\lambda + uD), \qquad \frac{d}{ds}\frac{1}{\rho u} = \frac{D}{2\rho}.$$

We compute return along increasing s on the transversal $s = 0$, so $\mathcal{R}(u_0) = u(L; u_0)$ for the solution starting at u_0. The reciprocal coordinate turns this calculation into a single period. Put

$$I_0 = \int_0^L \frac{D}{2\rho}\, ds.$$

Periodicity of ρ gives

$$\mathcal{R}(u) = \frac{u}{1 + \rho(0) I_0 u}. \tag{3.3}$$

Thus $\mathcal{R}'(0) = 1$ in every case, but the full return is the identity only if $I_0 = 0$.

There is also a transverse coordinate regular at the central leaf. Choose periodic ω with $\omega_s = D/(2\rho) - I_0/L$. Then

$$z = \frac{\rho u}{1 - \rho u \omega(s)} \tag{3.4}$$

has $z_u(s, 0) = \rho(s) > 0$ and changes the characteristic field to $\partial_s - (I_0/L) z^2 \partial_z$. When $I_0 = 0$, z is a smooth first integral across $u = 0$; on a one-sided band, $v = z^{-1}$ is more convenient.

Remark 3.2 (What the period measures). The form $(D/(2\rho))\, ds$ has one cohomological obstruction to a periodic primitive: its integral I_0. In this principal-normal class, $\mathcal{R}''(0) = -2\rho(0) I_0$, so the quadratic jet detects the whole return. Along the central curve, $\mathrm{II} = \left(\begin{smallmatrix} 0 & \tau \\ \tau & 0 \end{smallmatrix}\right)$ in the orthonormal basis (T, E). The transverse normal curvature is zero, which explains the automatic multiplier one. This does not say that the whole band is minimal.

3.3. **Classification of the supported kernel.** The *supported kernel* consists of the fields $y \in C_c^\infty((\mathbb{R}/L\mathbb{Z}) \times (0,b), \mathbb{R}^3)$ satisfying $S_X y = 0$.

Theorem 3.3 (Supported kernel on a ruled band)**.** *On a sufficiently thin one-sided band $0 < u < b$, a nonzero compactly supported infinitesimal bending exists exactly when $I_0 = 0$. In that case choose periodic ω with $\omega_s = D/(2\rho)$ and define*

$$v(s,u) = \frac{1}{\rho(s)u} - \omega(s), \qquad c_b = \max_s \left(\frac{1}{\rho(s)b} - \omega(s) \right).$$

Every supported bending has a unique expression

$$y_F = \frac{\tau u}{2\rho} F'(v) T + \left[uF(v) - \frac{1-ku}{2\rho} F'(v) \right] n, \qquad F \in C_c^\infty((c_b, \infty)). \tag{3.5}$$

Proof. Step 1: reduce the strain to transport. Write $y = \alpha T + \chi E + \beta n$. The uu equation is $\chi_u = 0$; compact support therefore gives $\chi = 0$. The remaining equations are

$$(1-ku)\alpha_s + u\tau\beta_s = 0, \qquad (1-ku)\alpha_u + u\tau\beta_u + k\alpha - \tau\beta = 0.$$

For $p = (1-ku)\alpha + u\tau\beta$, these are equivalent to

$$\alpha = p - \frac{u}{2} p_u, \qquad \beta = \frac{kp + (1-ku)p_u/2}{\tau}, \qquad p_s - \frac{u}{2}(\lambda + uD)p_u = -uDp. \tag{3.6}$$

Consequently $f = p/(\tau u^2)$ is constant along the nonruling characteristics.

Step 2: support forces identity return. Extend the field by zero to $u > 0$. A characteristic on which $f \neq 0$ cannot leave the compact support: transport preserves its nonzero value. It exists for all s and stays in a compact interval bounded away from $u = 0$. But $(\rho u)^{-1}$ gains I_0 on every circuit, which is incompatible with this boundedness unless $I_0 = 0$.

Step 3: reconstruct all supported fields. When $I_0 = 0$, the complete leaves in the band are

$$u_c(s) = \frac{1}{\rho(s)(c + \omega(s))}, \qquad c > c_b. \tag{3.7}$$

Taking $p = \tau u^2 F(v)$ in (3.6) gives (3.5). A compact interval of levels gives a compact subannulus, so every displayed field is supported. Conversely, if a field is supported in $u_- \le u \le u_+$ with $0 < u_- < u_+ < b$, every nonzero level satisfies

$$\max_s \left(\frac{1}{\rho(s)u_+} - \omega(s) \right) \le c \le \min_s \left(\frac{1}{\rho(s)u_-} - \omega(s) \right).$$

The left endpoint is greater than c_b and the right endpoint is finite. Evaluation of f at $s = 0$ therefore gives a smooth compactly supported function F on (c_b, ∞). Transport proves reconstruction, and every such level occurs on a complete leaf, proving uniqueness. □

We call a subband *saturated* for the nonruling foliation when it contains the complete leaf through each of its points. The saturated open annular subbands in the identity case are precisely $v^{-1}(J)$ for open intervals $J \subset (c_b, \infty)$. Disjoint transverse supports in (3.5) give disjoint surface supports. This feature, rather than just the dimension of the kernel, is what will permit exact independent sign choices.

Figure 2 gives a geometric reading of (3.5). A test function of v selects whole closed leaves, so its bending support is a union of complete leaves. Its derivative also enters the field: y_F is a coupled tangential and normal displacement, with no ruling component. The strain vanishes, but the quadratic metric correction need not vanish. Consequently the opposite signs give the common metric described in (1.1), rather than an isometric path through X.

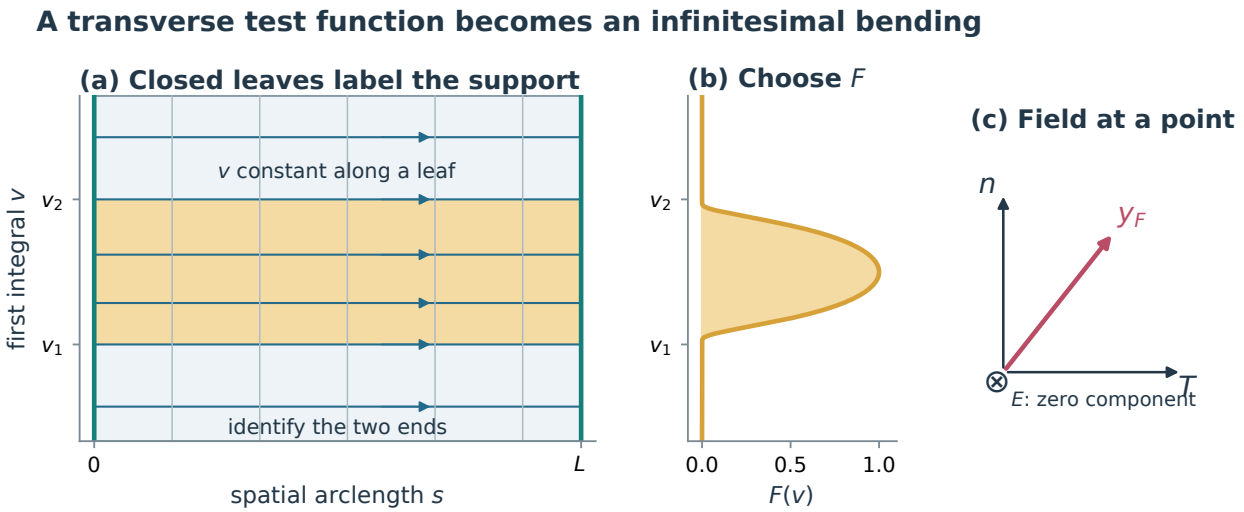


FIGURE 2. The bending map $F \mapsto y_F$ in Theorem 3.3. (a) Cutting a saturated subband at $s = 0$ gives a cylinder diagram with $s = 0$ and $s = L$ identified. Horizontal curves have constant first integral v; vertical curves correspond to rulings at fixed s. (b) For $c_b < v_1 < v_2$, a smooth function supported between v_1 and v_2 gives a field supported in the gold subannulus. (c) At each point the exact field has components $\langle y_F, T\rangle = \tau u F'(v)/(2\rho)$ and $\langle y_F, n\rangle = uF(v) - (1-ku)F'(v)/(2\rho)$, while $\langle y_F, E\rangle = 0$. The displayed arrow illustrates this decomposition; its length and angle are schematic. The characteristic transport equation, not the arrow, establishes the bending identity.

3.4. **The return multiplier and the Han–Khuri integral.** For later applications we record what the linear return measures on an arbitrary saddle surface.

Lemma 3.4 (Asymptotic return multiplier)**.** *Let Γ be an embedded closed asymptotic curve on an oriented surface with $K < 0$ near Γ. Parametrize it by oriented arclength, write T for its tangent and Z for a unit perpendicular tangent, and set*

$$k_g = \langle \nabla_T T, Z\rangle, \qquad k_n = \mathrm{II}(Z,Z), \qquad \sigma = \operatorname{sign} \mathrm{II}(T,Z).$$

Here ∇ and tangent inner products use the induced metric. If μ is the return multiplier for this orientation of Γ, then

$$\log \mu = \frac{\sigma}{2} \oint_\Gamma k_g k_n |K|^{-1/2}\, ds. \tag{3.8}$$

Proof. Choose Fermi coordinates (s,t) with metric $h^2 ds^2 + dt^2$ and $h(s,0) = 1$, and write $\mathrm{II} = L\,ds^2 + 2M\,ds\,dt + N\,dt^2$. On $t = 0$ we have $L = 0$, $N = k_n$, and $M = \sigma\sqrt{-K}$. Codazzi gives $L_t = M_s - k_g N$. If the asymptotic slope is $t_s = f(s,t)$, then $f(s,0) = 0$ and $L + 2Mf + Nf^2 = 0$ implies $f_t = -L_t/(2M)$ on the leaf. The variational equation yields

$$\log \mu = -\frac{1}{2}\oint \frac{L_t}{M}\, ds = -\frac{1}{2}\oint \frac{M_s}{M}\, ds + \frac{1}{2}\oint \frac{k_g N}{M}\, ds.$$

The first integral is zero by periodicity. Substitution of M and N in the second gives the assertion. □

The integral in [14] detects the linear multiplier. Equation (3.3) explains why multiplier one alone does not imply full identity return.

The identity-return mechanism also produces the visibly broader examples in Figure 3. In spherical arclength r, constant spatial speed $a = 1$ gives $I_0 = \frac{1}{2}\oint \kappa_r\, dr = 0$. Closure still requires the separate vector moment $\oint aP\, dr = 0$. These examples have different normal tracks from the

hemispherical seed used for the torus; they illustrate the return calculation without supplying its visibility and completion package.

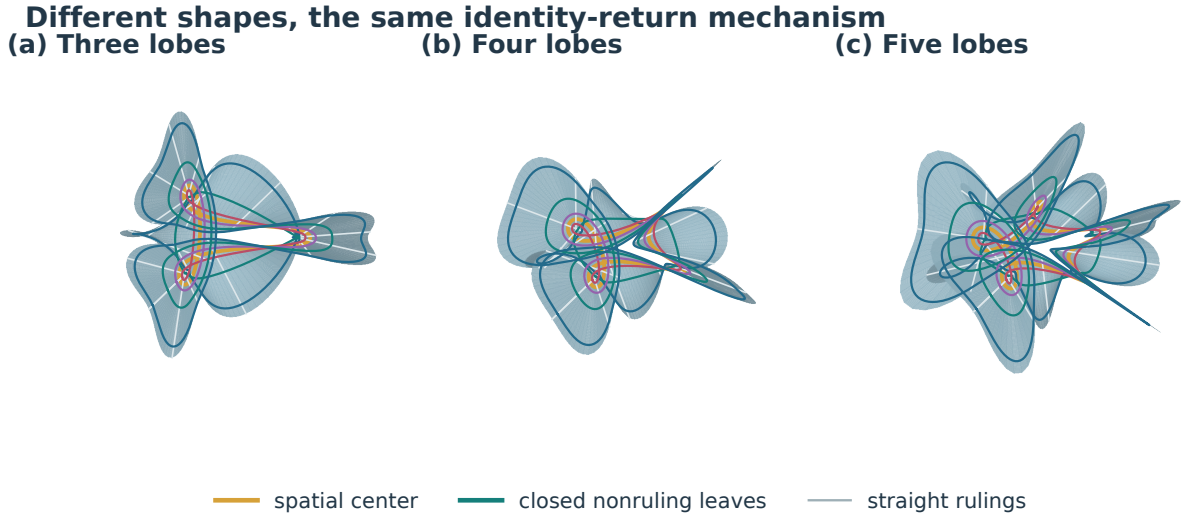


FIGURE 3. Three-, four-, and five-lobed constant-speed ruled bands from the numerical atlas. All axes in each model use equal physical units; the broad strips come from actual ruling widths, not a transverse stretch. Gold is the closed spatial center, colored curved traces show the closed nonruling asymptotic family, and pale straight segments show the ruling family. Traces are overlaid to make the foliation visible. With $a = 1$, the nonruling equation is $u_r = -\kappa_r u^2/2$, whose reciprocal gains zero over a period. These are illustrations by immersed bands: self-intersections are allowed in the saved models. They are not asserted to be embedded protected cores or completed tight tori.

4. Closing a normal loop and cancelling its return period

We now construct the frame used in Section 3. The input is a loop of normals, not a spatial curve. A positive speed determines the spatial primitive. Closure and return then become two explicit conditions on that speed, which can be solved separately.

4.1. **The two conditions on speed.** Let $\zeta : \mathbb{R}/\mathcal{L}\mathbb{Z} \to \mathbb{S}^2$ be a regular spherical loop parametrized by spherical arclength r. Define

$$P = -\zeta \times \zeta_r, \qquad \zeta_{rr} = -\zeta - \kappa P, \qquad P_r = \kappa \zeta_r. \tag{4.1}$$

These formulas fix the sign of its geodesic curvature κ. A smooth positive function a on $\mathbb{R}/\mathcal{L}\mathbb{Z}$ determines the spatial primitive $\Gamma_a(r) = x_0 + \int_0^r a(t)P(t)\,dt$. If $ds = a\,dr$, then the frame (P, ζ_r, ζ) has $k = \kappa/a$ and $\tau = -1/a$. Thus

$$\mathbf{M}(a) = \oint aP\,dr = 0, \qquad \mathscr{B}(a) = \oint \frac{\kappa_r}{\sqrt{a}}\,dr = 0, \qquad I_0 = \frac{1}{2}\mathscr{B}(a). \tag{4.2}$$

The first equation closes the spatial curve; the second cancels the full return period. We call a *closing* when $\mathbf{M}(a) = 0$ and *balanced* when both $\mathbf{M}(a) = 0$ and $\mathscr{B}(a) = 0$. Replacing $\Gamma_r = aP$ and $E = \zeta_r$ by $\Gamma_r = -aP$ and $E = -\zeta_r$ gives the positively oriented frame $(-P, -\zeta_r, \zeta)$ with $\tau = 1/a$ and the same conditions.

4.2. **A local speed correction that preserves moments.** The next lemma makes an arbitrarily small positional change while prescribing a possibly large nonlinear period. Its smallness is in L^1 for the speed, rather than in curvature or torsion norms.

Lemma 4.1 (Finite-moment period prescription)**.** *Let $T > 0$, and let $f : \mathbb{R}/T\mathbb{Z} \to \mathbb{R}^m$ and $g : \mathbb{R}/T\mathbb{Z} \to \mathbb{R}$ be smooth, with g taking both signs. All integrals and the L^1 norm below use dt over one period. For every smooth $b > 0$, $B_* \in \mathbb{R}$, and $\eta > 0$, there is a smooth $a > 0$ such that*

$$\|a - b\|_{L^1} < \eta, \qquad \oint af = \oint bf, \qquad \oint g/\sqrt{a} = B_*. \tag{4.3}$$

If $d = \dim \operatorname{span} f(\mathbb{R}/T\mathbb{Z})$, the correction uses at most $d + 1$ arbitrarily short intervals.

Proof. Choose a slowdown and compensating controls. If the period already equals B_*, take $a = b$. Otherwise choose a short interval K where g has the sign of $B_* - \oint g/\sqrt{b}$. Choose d disjoint control intervals O_j away from K, with smooth bumps ψ_j whose moments $\oint \psi_j f$ form a basis of $\operatorname{span} f(\mathbb{R}/T\mathbb{Z})$. One can first choose independent sample values of f, then a center for K in the required sign region away from those finitely many points, and finally shrink all intervals. Let $0 \le \chi \le 1$ be supported in K and equal one on a smaller arc. Solve

$$\sum_{j=1}^{d} c_j \oint \psi_j f = \oint \chi b f,$$

and set, for $0 \le t < 1$,

$$a_t = b - t\chi b + t \sum_{j=1}^{d} c_j \psi_j.$$

This path preserves every linear moment. The argument also covers $d = 0$, when there are no controls.

Keep the correction small and prescribe the period. Fixing the controls and shortening K gives $c_j = O(|K|)$. Thus a_t is positive on the controls and $\|a_t - b\|_{L^1} < \eta$, uniformly in t. On K, positivity follows from $t < 1$. On the plateau, $a_t = (1 - t)b$, so the contribution of $g/\sqrt{a_t}$ diverges with the chosen sign as $t \uparrow 1$. All contributions on K have that sign; those on the complement stay bounded. Continuity therefore attains B_* at some $t_* < 1$. Take $a = a_{t_*}$. □

4.3. **The structural existence criterion.**

Theorem 4.2 (Normal-loop existence criterion)**.** *For the loop in* (4.1)*, the following are equivalent:*
(i) *0 belongs to the ordinary three-dimensional interior of* $\operatorname{conv}\{P(r) : 0 \le r \le \mathcal{L}\}$*;*
(ii) *a smooth positive speed b satisfies $\mathbf{M}(b) = 0$;*
(iii) *a smooth positive speed a satisfies both $\mathbf{M}(a) = 0$ and $\mathscr{B}(a) = 0$.*
This characterizes closed immersed principal-normal identity-return bands with the prescribed normal loop. For every closing speed b, the balanced speed a can be chosen arbitrarily close to b in L^1.

Proof. From the convex hull to a closing speed. Assume (i) and consider

$$\Phi(\lambda) = \oint e^{\lambda \cdot P(r)}\, dr, \qquad \lambda \in \mathbb{R}^3.$$

Its Hessian is positive definite because P spans $\mathbb{R}^3$. The interior condition gives a uniform positive lower bound for $\max_r v \cdot P(r)$ as v ranges over unit vectors. Uniform continuity then gives arcs of uniformly positive length on which this scalar product is bounded below. Hence Φ is coercive. At its unique minimizer λ_*, the positive speed $b = e^{\lambda_* \cdot P}$ has $\mathbf{M}(b) = \nabla\Phi(\lambda_*) = 0$.

Why positive closure forces the interior condition. Assume (ii). If (i) fails, separation gives a nonzero v with $v \cdot P \ge 0$. Since $b > 0$ and the moment is zero, $v \cdot P = 0$ everywhere. Differentiation gives $\kappa v \cdot \zeta_r = 0$. On any interval where $\kappa \ne 0$, v is perpendicular to both P and ζ_r, so ζ is parallel

to v. This contradicts $|\zeta_r| = 1$. Thus $\kappa = 0$ everywhere, making P constant; a positive speed along a constant unit vector cannot close. This proves (i).

Cancel the period without changing closure. A closing speed also forces κ to be nonconstant. Indeed, if $\kappa = c$, then $C = P - c\zeta$ is constant and $P \cdot C = 1$, giving $\mathbf{M}(b) \cdot C = \oint b > 0$. Therefore κ_r takes both signs. Apply Lemma 4.1 with $f = P$, $g = \kappa_r$, and $B_* = 0$. This gives (iii) with arbitrary L^1 control; (iii) implies (ii) immediately. The frame formulas and (4.2) then produce the asserted immersed band, whose curvature is negative. □

Corollary 4.3 (Relative upgrade of an embedded seed)**.** *If a closing speed b has an embedded primitive Γ_b, the balanced primitive can be chosen embedded and arbitrarily uniformly close to it. The two are connected through embedded closed curves with the same oriented tangent and normal at each parameter value. If a fixed linear projection of Γ_b is a regular embedded curve, the same projection of the balanced primitive can be kept embedded. If ζ is embedded, the Gauss map of a sufficiently thin resulting band is injective.*

Proof. Use the same initial point for all primitives. Their uniform distance is bounded by the L^1 change of speed. On a short parameter arc, some fixed linear functional is strictly increasing along every positive-speed primitive of P; this excludes nearby collisions. Compact separation for Γ_b excludes all other collisions under a small uniform change. The argument holds along the whole correcting path and for a specified regular embedded projection. Tangent and normal directions are fixed. Finally the ruled Gauss map is locally invertible and injective on the compact central curve; it is injective on a thin neighborhood. □

For a unit tangent T and $Z = N \times T$, the *geodesic torsion* is $\tau_g = \mathrm{II}(T, Z)$. Along an asymptotic curve, $K = -\tau_g^2$, so an embedded closed asymptotic curve on an oriented $K < 0$ surface has nonzero geodesic torsion [13, §2.2] and a regular normal curve. Its normal, parametrized by spherical arclength, gives the frame above with one torsion sign. The corollary upgrades it to an identity-return ruled band without changing its knot type or normal self-linking. It does not deform the original surface isometrically. The global completion still needs the two visible traces supplied in the next section.

Remark 4.4 (The signed action of speed and curvature)*.* With $\rho = a^{-1/2}$, define the *return action* of the closed planar path $r \mapsto (\kappa(r), \rho(r))$ by

$$\mathscr{A} = \oint \rho \, d\kappa = 2I_0.$$

Its oriented algebraic area is $\frac{1}{2}\oint(\kappa \, d\rho - \rho \, d\kappa) = -\mathscr{A}$, for the standard orientation $d\kappa \wedge d\rho$. A speed law $a = A(\kappa)$, with A smooth and positive near the range of κ, cancels this action automatically, but need not solve the independent closing equation $\oint A(\kappa) P \, dr = 0$. The correction lemma can prescribe any return period while preserving closure; adding 1 to the linear moments also preserves total length. The positional control remains C^0, obtained from L^1 control of speed.

5. Constructing the embedded protected core

The structural criterion guarantees an immersed identity-return band. For completion we need more: embedded spatial and normal curves, an embedded horizontal projection, and two visible planar traces. The seed below supplies these together. Its long quantitative verification is deferred to Appendix B.

5.1. **Visibility and the seed package.**

For $|z| > R > 0$ define the unit vector

$$V_R(z) = \frac{Rz + \sqrt{|z|^2 - R^2}\, Jz}{|z|^2}. \tag{5.1}$$

Its numerator consists of perpendicular vectors with squared lengths summing to $|z|^4$.

Definition 5.1. Regular closed planar curves $p, \delta : \mathbb{R}/T\mathbb{Z} \to \mathbb{R}^2$, $T > 0$, with a common parameter are *visible at radius* R if

$$|\delta| > R, \qquad p' \cdot V_R(\delta) > 0, \qquad \delta' \cdot V_R(\delta) > 0. \tag{5.2}$$

These inequalities are required at every parameter value.

These conditions persist under common orientation-preserving reparametrization. On a compact parameter circle their strict margins also persist under small C^1 changes, or small positional changes with oriented tangent directions fixed. The connector construction will use visibility to replace a complicated gradient trace by a circle.

Identify $\mathbb{R}^2$ with $\mathbb{C}$. For an integer $N \geq 2$, set $\epsilon = N^{-1}$, $T_N = 2\pi/N$, and $\xi_N = e^{iT_N}$. Define

$$\beta_N(t) = (1 + \epsilon \sin Nt) \exp\big(i[t + 2\epsilon \sin Nt]\big). \tag{5.3}$$

Choose k_N as the unique root of

$$\int_0^{2\pi} e^{-k\cos x}(1 + \epsilon \sin x)^2(1 + 2\cos x)\, dx = 0, \tag{5.4}$$

and put

$$Z_N = \frac{1}{2\pi}\int_0^{2\pi} e^{-k_N \cos x}\, dx, \qquad m_N(t) = Z_N^{-1} e^{-k_N \cos Nt}.$$

The partner curve is defined by integrating a positive multiple of the same tangent, with an initial point chosen for rotational covariance:

$$I_N = \int_0^{T_N} m_N \beta_N'\, dt, \qquad \delta_N(t) = \frac{I_N}{\xi_N - 1} + \int_0^t m_N(s)\beta_N'(s)\, ds. \tag{5.5}$$

For any closed planar pair with a common parameter, we call $\oint \det(p, \delta')\, dt$ its *mixed action*.

Lemma 5.2 (Corrugated analytic seed)**.** *For every integer $N \geq 10^4$, β_N and δ_N are real analytic positively oriented embedded curves enclosing the origin. If Q_N is rotation through $2\pi/N$, then*

$$\beta_N(t + T_N) = Q_N \beta_N(t), \qquad \delta_N(t + T_N) = Q_N \delta_N(t), \qquad \delta_N' = m_N \beta_N', \quad m_N > 0, \tag{5.6}$$

and

$$\int_0^{2\pi} \det(\beta_N, \delta_N')\, dt = 0. \tag{5.7}$$

The pair (β_N, δ_N) is visible at $R_+ = 1/4$, and

$$(\widetilde{\beta}_N, \widetilde{\delta}_N)(t) = \big(A_* \delta_N(-t), A_* \beta_N(-t)\big)$$

is visible at $R_- = 4/5$. The spherical lift

$$\zeta_N(t) = \frac{(\beta_N(t), 1)}{\sqrt{1 + |\beta_N(t)|^2}}$$

is embedded in the northern hemisphere and its geodesic curvature takes both signs.

Proof. Appendix B verifies the root, covariance, action, embedding, visibility, and curvature assertions in that order. □

Remark 5.3 (The role of corrugation). The curves satisfy $\beta_N = e^{it} + O(N^{-1})$ and $\delta_N = (i/2)e^{it} + O(N^{-1})$ in position, while their common unit tangent has order-one oscillations. This allows signed angular backtracking and zero mixed action without losing nearly circular positional geometry. The latter makes both visibility estimates uniform. The exact threshold 10^4 is only a convenient sufficient bound.

The three views in Figure 4 separate the positional and tangent effects of corrugation. A modest value of N makes the global shape legible, whereas a uniformly enlarged single cell at $N = 10^4$ shows the small-scale motion within the sufficient analytic range. The partner is computed from a positive multiple of the same tangent; its closed primitive is a different curve, not a scaled copy of β_N.

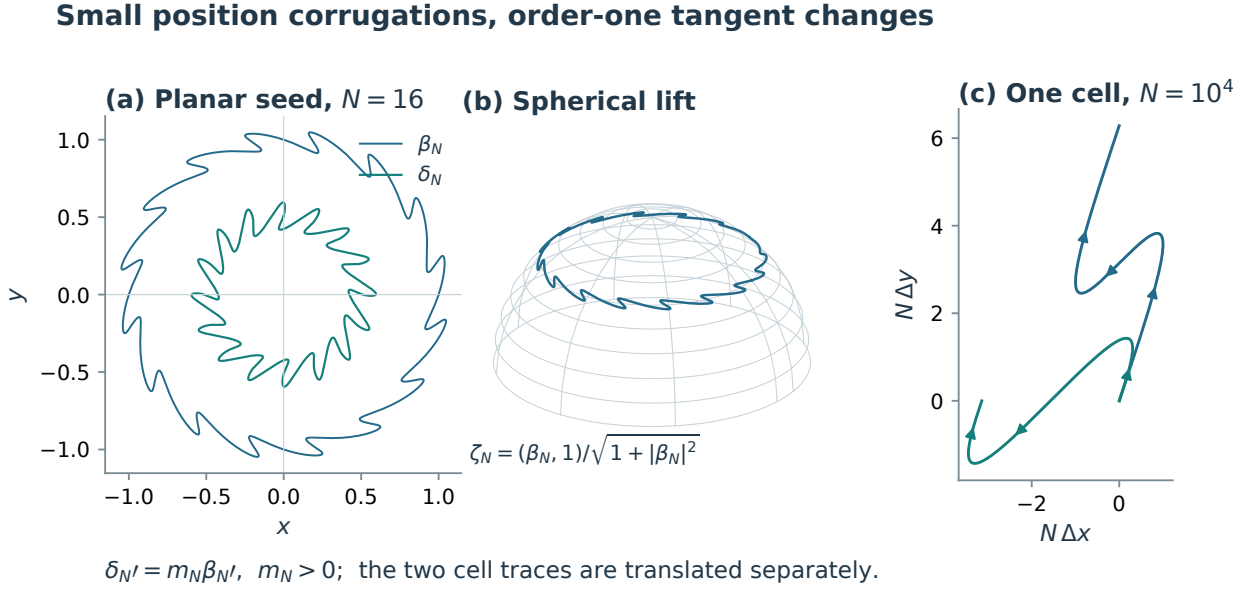


FIGURE 4. The seed formulas (5.3)–(5.5). (a) Blue is β_N and teal is δ_N at $N = 16$. (b) The corresponding northern spherical lift ζ_N. These two panels make corrugation visible; $N = 16$ is below the sufficient range of Lemma 5.2, so they are illustrations of the formulas. (c) One full cell at $N = 10^4$: each curve is translated to its own initial point, and both coordinate differences are multiplied by N. This is an isotropic magnification, not a common placement of the two curves. Arrows show the parameter direction. The oriented tangents agree at corresponding parameters because $\delta'_N = m_N \beta'_N$ with $m_N > 0$. The root defining m_N cancels the mixed action, while the covariant initial point closes the partner. The seed estimates are proved in Appendix B.

5.2. **From planar traces to a closed framed curve.** Fix one seed and suppress its subscript N. Let Q denote rotation by $2\pi/N$, extended to $\mathbb{R}^3$ by fixing the third coordinate. The spherical arclength is $r(t) = \int_0^t |\zeta_t(s)|\, ds$, and $\ell = r(T_N)$. Reparametrize ζ, β, δ, m by r. The coordinate transitions used in this section are summarized here:

t	original seed parameter, of period 2π
r	spherical arclength of ζ, of period $N\ell$
$a(r)$	positive spatial speed: $\Gamma_r = aP$
s	spatial arclength: $ds = a\, dr$, of period L
u	distance along the ruling $E = \zeta_r$
v	transverse first integral labelling closed leaves

Covariance gives $\zeta(r + \ell) = Q\zeta(r)$. Its frame is

$$P = -\zeta \times \zeta_r, \qquad \zeta_{rr} = -\zeta - \kappa P, \qquad P_r = \kappa \zeta_r. \tag{5.8}$$

Both κ and $|\beta|$ are ℓ-periodic. With $w = \sqrt{1+|\beta|^2}$, the planar expression is

$$P = \frac{(-J\beta_r, -\det(\beta, \beta_r))}{w^2}. \tag{5.9}$$

Define the spatial lift by

$$\Gamma_h = -J\delta, \qquad (\Gamma_3)_r = -\det(\beta, \delta_r), \tag{5.10}$$

where h denotes horizontal components. The vertical integrand is cell-periodic. Its total integral is zero by (5.7), which is invariant under the common reparametrization, so each of the N equal cell integrals is zero. Thus the vertical primitive is periodic on every cell. Fix its additive constant. Since $\delta_r = m\beta_r$, (5.9) gives

$$\Gamma_r = a_0 P, \qquad a_0 = mw^2 > 0. \tag{5.11}$$

Its horizontal projection is embedded, so Γ is embedded.

For any positive ℓ-periodic speed a, closure reduces to the scalar moment

$$M(a) = \int_0^\ell aP_3 \, dr = 0. \tag{5.12}$$

Indeed, the horizontal cell increments are successive rotations and sum to zero; the vertical increments are all equal. Choose the horizontal initial point by

$$(Q - I)\Gamma_h(0) = \int_0^\ell aP_h \, dr. \tag{5.13}$$

Then $\Gamma(r+\ell) = Q\Gamma(r)$. The new planar trace $\delta_a = J\Gamma_h$ satisfies $(\delta_a)_r = (a/w^2)\beta_r$: changing speed preserves its oriented tangent direction. The initial speed recovers δ and satisfies $M(a_0) = 0$.

The return balance over one cell is

$$\mathscr{B}_\ell(a) = \int_0^\ell \frac{\kappa_r}{\sqrt{a}} \, dr. \tag{5.14}$$

We use $\mathscr{B}_\ell$ here to distinguish this scalar from the support tensor B_H. For these ℓ-periodic speeds, closing and balance are equivalently $M(a) = 0$ and $\mathscr{B}_\ell(a) = 0$.

Proposition 5.4 (One-slowdown holonomy correction)**.** *If $b > 0$ is ℓ-periodic with $M(b) = 0$, κ is nonconstant, and $P_3 \not\equiv 0$, then for every $\eta > 0$ a positive smooth ℓ-periodic a satisfies*

$$\|a - b\|_{L^1(0,\ell)} < \eta, \qquad M(a) = 0, \qquad \mathscr{B}_\ell(a) = 0.$$

The change is supported in at most two arbitrarily short intervals.

Proof. Apply Lemma 4.1 on the cell circle with $f = P_3$, $g = \kappa_r$, and $B_* = 0$. The moment rank is one, and the periodic nonconstant κ has derivative of both signs. One slowdown and one compensating control suffice. □

Theorem 5.5 (Balanced embedded speed)**.** *For every $\eta_0 > 0$, the seed admits a positive smooth ℓ-periodic speed a with*

$$\|a - a_0\|_{L^1(0,\ell)} < \eta_0, \qquad M(a) = 0, \qquad \mathscr{B}_\ell(a) = 0. \tag{5.15}$$

For small enough η_0, its spatial primitive and horizontal projection are embedded, and both visibility conditions persist.

Proof. The seed has nonconstant κ. Also $P_3 \not\equiv 0$: otherwise $\det(\beta, \beta_r) = 0$ would prevent β from winding around zero. The proposition therefore gives the speed. For fixed N, the initial point in (5.13) varies continuously with the cell integral, and integration of aP gives uniform closeness of primitives. On short parameter arcs, a fixed linear functional is monotone along all positive-speed primitives; on the remaining pairs, the original embedded curve has positive separation. This proves embedding for the spatial curve and its horizontal projection. Visibility, after division by positive

speeds, depends on positions and the fixed oriented tangent directions. Its strict margins therefore survive the uniform positional change. The seed parameter N is fixed before choosing the speed correction. The band width is chosen only after the final positive speed a is fixed; the L^1 estimate does not give uniform bounds for its curvature or torsion. □

Remark 5.6 (Why a constant speed does not suffice). A constant a cancels the return period, but cannot close this hemispherical normal loop. Let $D \subset \{q_3 > 0\}$ be the disk bounded by ζ, with the sphere's outward orientation. Then

$$\oint \Gamma_r \, dr = -a \oint \zeta \times d\zeta = -2a \int_D \zeta \, dA.$$

The vertical component of the last integral is positive. Thus speed redistribution solves a real closure obstruction.

5.3. **The balanced band and its support tensor.** Fix the final speed and set $ds = a\,dr$, with $L = \int_0^{N\ell} a\,dr$. Set $\gamma(s) = \Gamma(r(s))$. In physical arclength the frame is $T = P$, $E = \zeta_r$, $n = \zeta$, and

$$k = \frac{\kappa}{a}, \qquad \tau = -\frac{1}{a}, \qquad \lambda = -\frac{a_r}{a^2}, \qquad D = \frac{\kappa_r}{a^2}. \tag{5.16}$$

These identities follow from $\partial_s = a^{-1}\partial_r$; in particular the subscripts defining λ and D are s derivatives. Hence

$$I_0 = \int_0^{N\ell} \frac{\kappa_r}{2\sqrt{a}}\, dr = \frac{N}{2}\mathscr{B}_\ell(a) = 0.$$

Theorem 3.3 supplies the localized kernel.

The ruled Gauss map is locally invertible because $K < 0$, and its central restriction is the embedded curve ζ. It is therefore injective on a thin band. Shrink the band so that its normals remain in the northern hemisphere. Horizontal projection is then locally invertible and is injective on the central curve, so it too is injective on a thin band. Fix a width satisfying both requirements.

Proposition 5.7 (Central support tensor). *In Gauss coordinates, the support tensor along the central curve is*

$$B_H = \begin{pmatrix} 0 & a \\ a & 0 \end{pmatrix} \tag{5.17}$$

in the spherical orthonormal basis (ζ_r, P). Its mixed entry is strictly positive.

Proof. The central derivative is $\Gamma_r = aP$, so $B_H\zeta_r = aP$. At $u = 0$, the ruled normal has $N_u = P/a$, while $X_u = \zeta_r$. Consequently $B_H P = a\zeta_r$, giving the second column. □

6. Positive exits without changing the bending region

The ruled core has closed characteristics, whereas the completion needs positive noncharacteristic seams. We create those seams in two collars where all retained bendings vanish. The core itself stays fixed.

6.1. Changing return and finding a positive curve.

Lemma 6.1 (Exit perturbation). *Let a saddle support in the northern hemisphere have a positively oriented embedded closed asymptotic curve enclosing the north pole. Suppose its return germ is the identity and the mixed support entry, evaluated on unit tangent and outward conormal, is positive. Inside any prescribed collar, an arbitrarily small C^2 perturbation fixing H and dH on the curve can make return attracting ($0 < \mu < 1$) or repelling ($\mu > 1$) in its positive circuit direction. The attracting choice gives arbitrarily close positive closed curves on the exterior side; the repelling choice gives them on the interior side.*

Proof. Step 1: express the multiplier through the normal second jet. Parametrize the curve $q_0(r)$ by spherical arclength, of period $\mathcal{L}$, and let $P = -q_0 \times (q_0)_r$ be its outward conormal. The Fermi map and metric are

$$q(r,t) = \cos t\, q_0(r) + \sin t\, P(r), \qquad h^2 dr^2 + dt^2, \qquad h = \cos t + \kappa(r) \sin t.$$

Here $t > 0$ is exterior and $(q_0)_{rr} = -q_0 - \kappa P$. Write $B_H = L\, dr^2 + 2M\, dr\, dt + N\, dt^2$, with $(L, M, N)|_{t=0} = (0, a, c)$ and $a > 0$. The coordinate expressions are

$$L = H_{rr} - \frac{h_r}{h} H_r + h h_t H_t + H h^2, \quad M = H_{rt} - \frac{h_t}{h} H_r, \quad N = H_{tt} + H.$$

On the seam they give

$$L_t = a_r + \kappa c. \tag{6.1}$$

For clarity, $a_r = H_{rrt} - \kappa_r H_r - \kappa H_{rr}$ there; subtracting this from L_t leaves $\kappa(H_{rr} + \kappa H_t + H) + \kappa(H_{tt} + H) = \kappa c$. The asymptotic slope vanishing on the seam is

$$t_r = f(r,t) = -\frac{L}{M + \sqrt{M^2 - LN}}, \qquad f_t(r,0) = -\frac{a_r + \kappa c}{2a}.$$

Integrating its variational equation gives

$$\mu = \exp\left(-\frac{1}{2} \int_0^{\mathcal{L}} \frac{\kappa c}{a}\, dr \right), \tag{6.2}$$

because the period of a_r/a is zero.

Step 2: change that jet while retaining the curve. Choose a cutoff $\chi(t)$ supported in the prescribed collar and equal one near zero. Put

$$H_\epsilon = H + \frac{\epsilon}{2} a(r) \kappa(r) t^2 \chi(t).$$

The first jet is unchanged; $L = 0$ and $M = a$ remain, while c changes to $c + \epsilon a \kappa$. Since the original multiplier is one,

$$\mu_\epsilon = \exp\left(-\frac{\epsilon}{2} \int_0^{\mathcal{L}} \kappa^2\, dr \right). \tag{6.3}$$

The integral is positive: otherwise the curve would be a great circle, which cannot lie in an open hemisphere. Thus $\epsilon > 0$ gives attraction and $\epsilon < 0$ repulsion. For fixed collar and cutoff, small ϵ preserves the saddle determinant by C^2 continuity.

Step 3: produce a positive closed graph on the chosen side. Fix such a nonzero ϵ, set $b(r) = f_t(r,0)$ for the perturbed support, and let $\bar{b} = (\log \mu_\epsilon)/\mathcal{L}$. The equation $\psi_r = (b - \bar{b})\psi$ has a positive periodic solution because $b - \bar{b}$ has zero mean. For a small real d, take the graph $t = d\psi(r)$. Uniform Taylor expansion gives

$$B_{H_\epsilon}(\partial_r + t_r \partial_t, \partial_r + t_r \partial_t) = -2abd\psi + 2ad\psi_r + O(d^2) = -2a\bar{b} d\psi + O(d^2).$$

This is positive for small $d > 0$ in the attracting case and small $d < 0$ in the repelling case. The graphs are embedded closed curves in the collar and converge smoothly to the seam. □

6.2. **Fixing the protected annulus and both exits.** Choose $I = (c_-, c_+)$ with compact closure in the complete-leaf interval of Theorem 3.3. Let V_0 be the compact annulus swept out by the levels $c \in [c_-, c_+]$. All fields from $C_c^\infty(I)$ are supported in V_0. Choose levels $c_b < c_1 < c_- < c_+ < c_2$. Their leaves bound a larger annulus, and their disjoint collars can be chosen separated from a neighborhood of V_0. Every retained bending vanishes on those collars.

In the Gauss domain, the leaves are nested positively oriented curves around the north pole. Their mixed entries remain positive by (5.17) and thinness. Apply the exit lemma with attraction on the outer leaf and repulsion on the inner leaf. Choose the perturbations small enough to preserve

embedding of the support immersion and its horizontal projection; then choose the positive graphs sufficiently close to the perturbed seams.

Let G be the planar potential of the perturbed support. Parametrize the resulting exit curves counterclockwise by p_-, p_+ and set $\gamma_\pm = \nabla G \circ p_\pm$. These paired traces, which we call the *positive exits*, satisfy

$$p'_\pm \cdot \gamma'_\pm > 0. \tag{6.4}$$

Both source and gradient maps are embeddings on a neighborhood of the annulus between them. The individual traces are positively oriented and enclose zero. The outer pair $(p_+, J\gamma_+)$ is visible at R_+; the inner pair needed after duality is

$$\big(S\gamma_-(-s), JSp_-(-s)\big), \tag{6.5}$$

visible at R_-. The central source and gradient for the final speed are β and $-J\delta_a$, where $\delta_a = J\Gamma_h$. Theorem 5.5 preserves visibility of (β, δ_a) and $(A_*\delta_a(-r), A_*\beta(-r))$. These strict inequalities persist on sufficiently close exit traces, and $SJ = -JS$ gives the reflected pair in (6.5).

The source order is p_- inside p_+. The gradient order is reversed: γ_+ lies inside γ_-. Indeed, the gradient is an orientation-reversing diffeomorphism, while both gradient boundary curves are parametrized counterclockwise; the induced boundary orientations determine which one is inner. Thus the minus sign in a trace name refers to its source boundary, not to its gradient position. We fix all these choices in order: final speed, thin band, protected levels and collars, small exit perturbations, then close positive exit graphs. Every required margin is strict on a compact set. From this point onward, an open neighborhood of V_0 is kept unchanged.

7. Support completion by dual radial filling

The protected band has two positive noncharacteristic exit curves. The only nonradial extension tool we retain is a visible connector: it makes an outgoing *gradient* trace circular. A differential Legendre transform then makes that circle a *source* seam, where a large radial quadratic term controls the saddle Hessian. On one end we continue to a polar circle; on the other we attach a square-root Legendre profile which becomes a reflectable parabolic neck. The same radial lemma treats both ends, but the roles of source and gradient are exchanged at each Legendre transform. We record those domains explicitly in the final completion proof. All changes to the support occur outside the protected core. A single ambient normalization, applied also to the retained fields, is recorded in Section 8.

Figure 5 follows the completion in two complementary views. The physical windows retain the actual band and its attachments; the domain diagram records which circular trace is a gradient and which is a source. At the final common physical scale the band and neck are microscopic, so the global cutaway serves to show the assembly, while the local windows show the attachment geometry.

7.1. **Relative smoothing at a positive seam.**

Lemma 7.1 (Relative saddle smoothing)**.** *Let U be a tubular neighborhood of a compact embedded closed planar seam Γ, divided into two sides by Γ. Let G_+ and G_- be smooth saddle potentials on neighborhoods of the respective closed sides, with matching values and gradients on Γ. If their common tangential Hessian is positive on Γ, the piecewise potential on U admits a smooth saddle interpolation, equal to it outside any prescribed neighborhood of Γ and arbitrarily close in C^1.*

Proof. The allowed second jets form a convex interval. Use tubular coordinates (s, t) with seam $t = 0$ and denote the piecewise function by G_{pw}. Matching first jets makes it C^1 with locally Lipschitz first derivatives. Differentiating the common gradient along the seam shows that the tangential and mixed Hessian entries agree. In a tangent-normal orthonormal frame the limiting Hessians are

$$H_\pm(s) = \begin{pmatrix} a(s) & b(s) \\ b(s) & c_\pm(s) \end{pmatrix}, \qquad a > 0, \quad c_\pm < b^2/a.$$

There is a uniform strict margin, and every convex combination remains indefinite because only the normal entry varies.

Mollify and retain the old pieces away from the seam. Extend the coordinate cylinder periodically in s and convolve with a symmetric nonnegative mollifier of radius δ. Since the gradient has no jump, the distributional Hessian has no seam measure. Close to the seam, the mollified Hessian is a weighted average of nearby one-sided Hessians, uniformly $o(1)$ from the displayed convex segments. Conversion from coordinate Hessians to Euclidean Hessians adds only $o(1)$ errors, since the gradients converge uniformly. Thus the determinant stays negative. Away from the seam, ordinary C^2 convergence gives the same conclusion.

Patch the mollified function to G_{pw} with a cutoff equal one for $|t| \le h$ and zero for $|t| \ge 2h$, where $h = \sqrt{\delta}$. On the transition strips the mollifier stays on one smooth side. Its value and gradient errors are $O(\delta^2)$ and $O(\delta)$. The additional Hessian terms are therefore $O(\delta^2 h^{-2} + \delta h^{-1}) = o(1)$. The saddle margin survives, the C^1 error tends to zero, and the old pieces are unchanged for $|t| \ge 2h$. Taking δ small places all changes inside the prescribed neighborhood. □

When using this lemma we cut the retained boundary inside an unchanged incoming subcollar. The unchanged boundary images then give global injectivity by the degree criterion.

7.2. **A connector with circular outgoing gradient.** The design is to draw a family of straight source segments from p, choose their lengths so the terminal gradient is a circle, and then use degree to rule out overlaps in both source and gradient.

Lemma 7.2 (From a visible trace to a circular gradient)**.** *Let $p, \delta : \mathbb{R}/T\mathbb{Z} \to \mathbb{R}^2$ be positively oriented embedded curves enclosing zero. Put $\gamma = -J\delta$ and assume*

$$p' \cdot \gamma' > 0, \qquad g' = \gamma \cdot p' \tag{7.1}$$

for a periodic g. If (p, δ) is visible at $R > 0$, an exterior source annulus carries a saddle potential with incoming value g and gradient γ, and outgoing gradient a positively parametrized circle of radius R. Both boundaries are positive for the Hessian. The source and gradient maps are diffeomorphisms of interiors onto their respective planar annuli. If the terminal gradient is written Re_θ, its terminal source trace $P_{\mathrm{out}}(\theta)$ also satisfies $P_{\mathrm{out}} \cdot e_\theta > 0$ for a sufficiently small connector parameter. If an incoming saddle potential supplies the data on a collar, relative smoothing leaves it unchanged on a subcollar away from the incoming seam and retains the outgoing germ.

Proof. Step 1: choose the segment directions and lengths. Set $\kappa_0 = \sqrt{|\delta|^2 - R^2}$ and choose an angular lift with

$$e_\varphi = V_R(\delta), \qquad \delta = Re_\varphi - \kappa_0 J e_\varphi. \tag{7.2}$$

Differentiation gives $\varphi' = \delta' \cdot e_\varphi / \kappa_0 > 0$. The argument of δ differs from φ by the periodic angle $-\arctan(\kappa_0/R)$, so $\varphi(s+T) = \varphi(s) + 2\pi$. Choose a small constant $\eta > 0$, put $\psi = \varphi - \eta$ and $w = \delta - Re_\psi$, and define

$$E_0 = \delta \cdot e_\psi - R, \quad A_0 = \det(w, p'), \quad B_0 = \det(w, \delta'), \quad C_0 = E_0 R \psi', \quad \ell = A_0/C_0. \tag{7.3}$$

All four scalars are positive for small η, uniformly in s: $E_0 = \kappa_0 \sin\eta - R(1-\cos\eta)$, $A_0 \to \kappa_0 p' \cdot e_\varphi > 0$, and $B_0 \to \kappa_0 \delta' \cdot e_\varphi > 0$.

Step 2: verify local potential and saddle determinant. On $0 \le u \le \ell(s)$ put

$$P(s,u) = p + uw, \qquad \Delta = A_0 + u(B_0 - C_0), \qquad \widehat{G}(s,u) = g + u\gamma \cdot w. \tag{7.4}$$

At the endpoints $\Delta = A_0$ and $\Delta = \ell B_0$, hence $\Delta > 0$. Since $\det(w, w') = B_0 - C_0$,

$$\det(P_s, P_u) = -\Delta < 0. \tag{7.5}$$

In a local inverse chart $G = \widehat{G} \circ P^{-1}$ has gradient

$$Y = \gamma + \frac{uB_0}{\Delta} Jw, \qquad Y_u = \frac{A_0 B_0}{\Delta^2} Jw. \tag{7.6}$$

Indeed, $Y \cdot P_u = \widehat{G}_u$; also $Jw \cdot P_s = \Delta$ and $\gamma' \cdot w = B_0$, so $Y \cdot P_s = \widehat{G}_s$. The pulled-back Hessian has uu entry zero and mixed entry $A_0 B_0/\Delta$. Dividing its determinant by the square of the source Jacobian gives

$$\det D^2 G = -\frac{A_0^2 B_0^2}{\Delta^4} < 0. \tag{7.7}$$

At $u = \ell$ the fraction in (7.6) is one, hence

$$Y(s, \ell(s)) = -RJe_{\psi(s)}. \tag{7.8}$$

Step 3: prove global source injectivity. As $\eta \downarrow 0$, convergence in each fixed smooth norm gives

$$\eta\ell \longrightarrow z_0 := \frac{p' \cdot e_\varphi}{R\varphi'} > 0, \qquad \eta P(s, \ell(s)) \longrightarrow -z_0 \kappa_0 J e_\varphi. \tag{7.9}$$

For $\theta = \psi - \pi/2$, the same limit gives $\eta P(s, \ell(s)) \cdot e_\theta \longrightarrow z_0 \kappa_0 > 0$. This is the radial-derivative condition needed after Legendre duality. The limit is a radial graph with positive radius and strictly increasing polar angle $\varphi - \pi/2$. Its radius and angular derivative have positive minima. For small η the terminal source curve is therefore embedded, positively oriented, and star-shaped, and its unscaled radius tends uniformly to infinity. It encloses p. The induced orientations of the two boundary curves are opposite. Lemma 2.4, together with (7.5), gives global injectivity of P and a single well-defined potential on its source annulus.

Step 4: check the seams and global gradient injectivity. Differentiating (7.9) and pairing with e_ψ yields

$$\eta \frac{d}{ds} P(s, \ell(s)) \cdot e_\psi \longrightarrow z_0 \kappa_0 \varphi' > 0.$$

The outgoing gradient derivative is $R\psi' e_\psi$, so its tangential Hessian is positive. The incoming one is positive by hypothesis. The incoming gradient curve $\gamma = -J\delta$ encloses the disk of radius R: it is embedded, surrounds zero, and stays outside that disk. Thus the gradient boundary curves are disjoint and nested. Equation (7.7) and the degree criterion give global gradient injectivity. Finally the incoming first jet is exactly (g, γ). Relative saddle smoothing attaches the incoming potential while retaining an unchanged incoming subcollar away from the seam and the outgoing germ. The boundary images still provide the same degree argument on the joined annulus. □

7.3. **Quadratic filling of a circular source seam.** The next lemma replaces the separate straightening, logarithmic, and cap constructions. Its proof uses two *linear* inequalities within the nonconvex saddle condition. For a function in polar coordinates,

$$D^2 f\big|_{(e_\theta, Je_\theta)} = \begin{pmatrix} f_{rr} & f_{r\theta}/r - f_\theta/r^2 \\ f_{r\theta}/r - f_\theta/r^2 & (f_{\theta\theta} + r f_r)/r^2 \end{pmatrix}. \tag{7.10}$$

Thus $f_{rr} < 0$ and $f_{\theta\theta} + r f_r > 0$ imply $\det D^2 f < 0$, regardless of the mixed derivative.

Lemma 7.3 (Quadratic filling at a circular seam)**.** *Let f be a smooth saddle potential on an exterior collar of $\{|y| = R\}$, with boundary traces*

$$h(\theta) = f(Re_\theta), \qquad b(\theta) = f_r(Re_\theta).$$

Assume

$$b(\theta) > 0, \qquad h''(\theta) + Rb(\theta) > 0. \tag{7.11}$$

There is a smooth saddle extension into $0 < |y| < R$, agreeing with f outside any prescribed seam neighborhood, and equal near the puncture to

$$f(y) = c|y| - \frac{M}{2}|y|^2 + d, \qquad c = MR, \tag{7.12}$$

where $M > 0$ can be arbitrarily large. If additionally the trace $\nabla f(Re_\theta)$ is a positively oriented Jordan curve enclosing zero, the extension can be chosen with globally injective gradient on the punctured disk joined to a sufficiently thin unchanged exterior collar, and the radial inner-gradient circle can be made to enclose that Jordan curve.

Proof. Choose a smooth function $\chi : [0, R] \to [0, 1]$ equal to zero for $r \le R/2$ and one for $r \ge 3R/4$. On $0 < r \le R$ put

$$f_M(r, \theta) = -\frac{M}{2}(r - R)^2 + \chi(r)\big[h(\theta) + (r - R)b(\theta)\big]. \tag{7.13}$$

At $r = R$ this has exactly the prescribed value and first derivatives. All the angular terms and their derivatives are bounded independently of M, while

$$(f_M)_{rr} = -M + O(1), \qquad (f_M)_{\theta\theta} + r(f_M)_r = Mr(R - r) + O(1). \tag{7.14}$$

At $r = R$, the full second expression is $h'' + Rb > 0$ by (7.11). By continuity and the nonnegative leading term, it remains positive in a fixed narrow collar below R for large M. Off that collar the leading term $Mr(R - r)$ is positive; on $r < R/2$ the angular error vanishes altogether. Hence both strict inequalities preceding the lemma hold throughout the punctured disk. Near $r = 0$ formula (7.13) is exactly (7.12) with $d = -MR^2/2$.

The two potentials have a common first jet on $r = R$, indefinite Hessians, and positive common tangential Hessian $(h'' + Rb)/R^2$. Lemma 7.1 gives a smooth interpolation confined to an arbitrarily small seam collar, leaving f unchanged on an exterior subcollar and retaining the radial inner germ. First fix M, then take that collar sufficiently small. The smoothing preserves negative Hessian determinant and positive tangential quadratic form; it need not preserve the sufficient inequality $(f_M)_{rr} < 0$ on the incoming side of the seam.

For the injectivity assertion, fix a nearby exterior circle $r = R_{\text{ext}} > R$ in the original collar. Its gradient trace P_{ext} is still a positively oriented Jordan curve enclosing zero. Choose M so large that $MR/2 > \max |P_{\text{ext}}|$, and confine the smoothing inside this exterior circle and outside $r = R/2$. For every $0 < r_0 < R/2$ the inner circle has gradient

$$\nabla f_M(r_0 e_\theta) = (MR - Mr_0)e_\theta.$$

This circle surrounds P_{ext}. Between the two boundaries $\det D^2 f < 0$, the boundary maps are embedded and nested, and their induced winding numbers are opposite. The annular degree criterion, Lemma 2.4, gives global gradient injectivity, also after relative smoothing. Letting $r_0 \downarrow 0$ gives an injective gradient on the entire joined punctured annulus; restriction gives injectivity on every smaller annulus. □

7.4. **A radial neck adapter.** The quadratic germ can be ended at the puncture, or it can be joined radially to a square-root germ. The latter has a particularly simple inverse Legendre transform.

Lemma 7.4 (Square-root neck adapter)**.** *Suppose a radial saddle potential $f = f(r)$ has $f' > 0$ and $f'' < 0$ near a circle $r = r_j > 0$. Choose any $0 < a < r_j$ and set*

$$\ell = r_j - a, \quad B = f'(r_j)^2\ell, \quad C = f(r_j) - 2f'(r_j)\ell. \tag{7.15}$$

Then

$$f_{\text{neck}}(r) = C + 2\sqrt{B(r - a)}, \qquad a < r \le r_j, \tag{7.16}$$

matches the value and radial derivative of f at r_j, and satisfies $f'_{\mathrm{neck}} > 0$, $f''_{\mathrm{neck}} < 0$. Relative radial smoothing gives a smooth saddle continuation, unchanged near its original incoming collar and exactly (7.16) *near $r = a$. The radial gradient there has norm tending to infinity as $r \downarrow a$.*

Proof. The derivative is $\sqrt{B/(r-a)}$, and the second derivative is $-\sqrt{B}/(2(r-a)^{3/2})$. The constants in (7.15) give the matching first jet. In a radial interpolation, the second derivative may be smoothed through negative functions while the first derivative remains positive; this is the one-dimensional special case of Lemma 7.1. The radial Hessian eigenvalues f'' and f'/r keep opposite signs. On the square-root segment the radial gradient is strictly monotone and becomes unbounded at a. □

7.5. **The two-sided extension theorem.** We now state explicitly what the global construction takes as input. Neither the identity-return property nor a ruling is used in this step. They were needed to create the protected band and its bendings, not to extend a positive boundary jet.

Proposition 7.5 (Two-sided support completion)**.** *Let G be a saddle potential on a neighborhood of a compact planar smooth annulus $\mathcal{A}$, with injective gradient there. Let $p_\pm : \mathbb{R}/T_\pm\mathbb{Z} \to \mathbb{R}^2$, $T_\pm > 0$, be embedded counterclockwise parametrizations of its inner and outer boundaries, and put $\gamma_\pm = \nabla G \circ p_\pm$. Assume the source boundary curves and the gradient boundary curves are nested Jordan curves enclosing zero, that $\gamma_\pm$ are counterclockwise, and that $p'_\pm \cdot \gamma'_\pm > 0$ everywhere. Assume $(p_+, J\gamma_+)$ is visible at some $R_+ > 0$ and $\big(S\gamma_-(-s), JSp_-(-s)\big)$ is visible at some $R_- > 0$. For any compact subannulus $V \Subset \mathcal{A}^\circ$, there is a saddle potential $\widetilde{G}$ on $\mathbb{R}^2 \setminus \{0\}$ equal to G on a neighborhood of V, with*

$$\begin{aligned} \widetilde{G}(p) &= R_N|p| - \frac{\mu}{2}|p|^2 + d_0 \quad (0 < |p| \ll 1), \\ \widetilde{G}(p) &= A|p| - \frac{B}{|p|} + d_\infty \qquad (|p| \gg 1), \end{aligned} \tag{7.17}$$

where $R_N > A > 0$ and $\mu, B > 0$. The gradient is a diffeomorphism

$$\nabla\widetilde{G} : \mathbb{R}^2 \setminus \{0\} \longrightarrow \{y \in \mathbb{R}^2 : A < |y| < R_N\}. \tag{7.18}$$

Every smoothing takes place outside the protected region.

Proof. At each step we take the Legendre transform of the final smoothed potential using its global gradient inverse. The unchanged collars identify successive potentials on open overlaps.

The exterior end. Apply Lemma 7.2 to $(p_+, J\gamma_+)$, retaining an incoming collar. Its terminal gradient is R_+e_θ, and the terminal source trace $P(\theta)$ is a positively oriented Jordan curve with $P \cdot e_\theta > 0$. Apply the differential Legendre transform $f = G^*$ on the gradient image. Its source seam is the circle $y = R_+e_\theta$; write $h(\theta) = f(R_+e_\theta)$. Then

$$f_r(R_+e_\theta) = P \cdot e_\theta > 0, \qquad h'' + R_+f_r = R_+P_\theta \cdot Je_\theta > 0.$$

The last inequality is precisely positivity of the connector's outgoing seam. Lemma 7.3 therefore continues f radially inward, with injective gradient. Stop within the quadratic radial region at $r = r_j$, and apply Lemma 7.4; choose its inner radius $a = A > 0$. The gradient of this dual continuation tends radially to infinity at $r = A$. Inverse Legendre transformation consequently extends the original potential toward $|p| = \infty$. Indeed, on the terminal square-root germ $|p| = f'_{\mathrm{neck}}(r)$, so $r = A + B/|p|^2$ and

$$p \cdot y - f_{\mathrm{neck}}(y) = A|p| - B/|p| - C.$$

This is the second germ in (7.17). For clarity, the three domains in this exterior construction are p-space for G, $y = \nabla G(p)$-space for $f = G^*$, and $p = \nabla f(y)$-space when we return to the original potential. Moving inward in the y-source annulus makes the p-gradient circles expand. The square-root endpoint $|y| = A$ therefore becomes the unbounded original-source end. Each relative smoothing retains an open incoming collar and the exact terminal radial germ.

The interior end. Use reflected differential duality: put $K(x) = G^*(Sx)$. In terms of the original inner traces (p_-, γ_-), the new source and gradient traces, with reversed parameter, are respectively

$$p^-(s) = S\gamma_-(-s), \qquad \gamma^-(s) = Sp_-(-s).$$

The action period still vanishes, since $\oint p_- \cdot d\gamma_- = -\oint \gamma_- \cdot dp_- = 0$. The original gradient curve γ_- is the outer boundary of the gradient annulus, so $S\gamma_-$ is the outer source boundary for K. Reflection reverses planar orientation and $s \mapsto -s$ reverses it again; both new trace parametrizations are therefore positive. The connector extends K on its exterior side. After taking K^*, this added region lies on the bounded source side of the original p_- boundary. This is the interior end we need. Its trace meets the same connector hypotheses by (6.5). Run the connector, then take $f = K^*$ and use Lemma 7.3 down to the puncture. On the unchanged initial collar

$$K^*(y) = G(Sy),$$

so the inverse reflection recovers the original G. The radial quadratic germ of K^* is invariant under S, giving the first germ in (7.17), with $\mu = M$ and $R_N = MR_-$. The coefficient M can be made arbitrarily large; we choose R_N larger than A and than the bounded intermediate gradient traces.

Global geometry and periods. We apply the degree criterion to different maps at different stages. First, the connector source map P has an embedded large outer boundary enclosing its incoming source curve; its negative Jacobian therefore proves source injectivity. Second, on the retained support plus connector, the gradient has nested boundary images: the retained original gradient boundary and the new radius-R circle. Degree gives a global inverse on this joined annulus, so the first dual potential is single-valued.

Third, in each dual filling, retain the far incoming boundary in an unchanged collar. The newly added small source circle is inside it, whereas its gradient is a large circle enclosing the incoming gradient trace. The coefficient in Lemma 7.3 is chosen after those traces are fixed. Degree makes the continued dual gradient injective on every such compact annulus. The radial neck has strictly monotone, expanding gradient radii, so the same argument persists down to any circle above its inner endpoint. These are the global inverse domains used to transform back. Relative smoothing changes no retained boundary germ in these degree applications.

Finally, the original inner and outer extensions agree with G on open collars on their respective sides. They thus define one potential $\widetilde{G}$ on the punctured plane. Its global gradient image is verified separately by the following exhaustion. On the inner and outer radial collars, for sufficiently small $\varepsilon > 0$ and large L,

$$\nabla\widetilde{G}(\varepsilon e_\theta) = (R_N - \mu\varepsilon)e_\theta, \qquad \nabla\widetilde{G}(Le_\theta) = (A + B/L^2)e_\theta.$$

The boundary circles are disjoint and nested, and the Hessian Jacobian is negative on the intervening source annulus. Lemma 2.4 proves that its gradient is a diffeomorphism onto the intervening planar annulus. Exhausting the punctured plane establishes (7.18). All Legendre transformations are genuine differentials of globally injective gradients on the relevant compact annuli; they therefore introduce no missing periods. The inner and outer potentials recover the original value and gradient on unchanged overlaps, so they assemble into one globally defined support. □

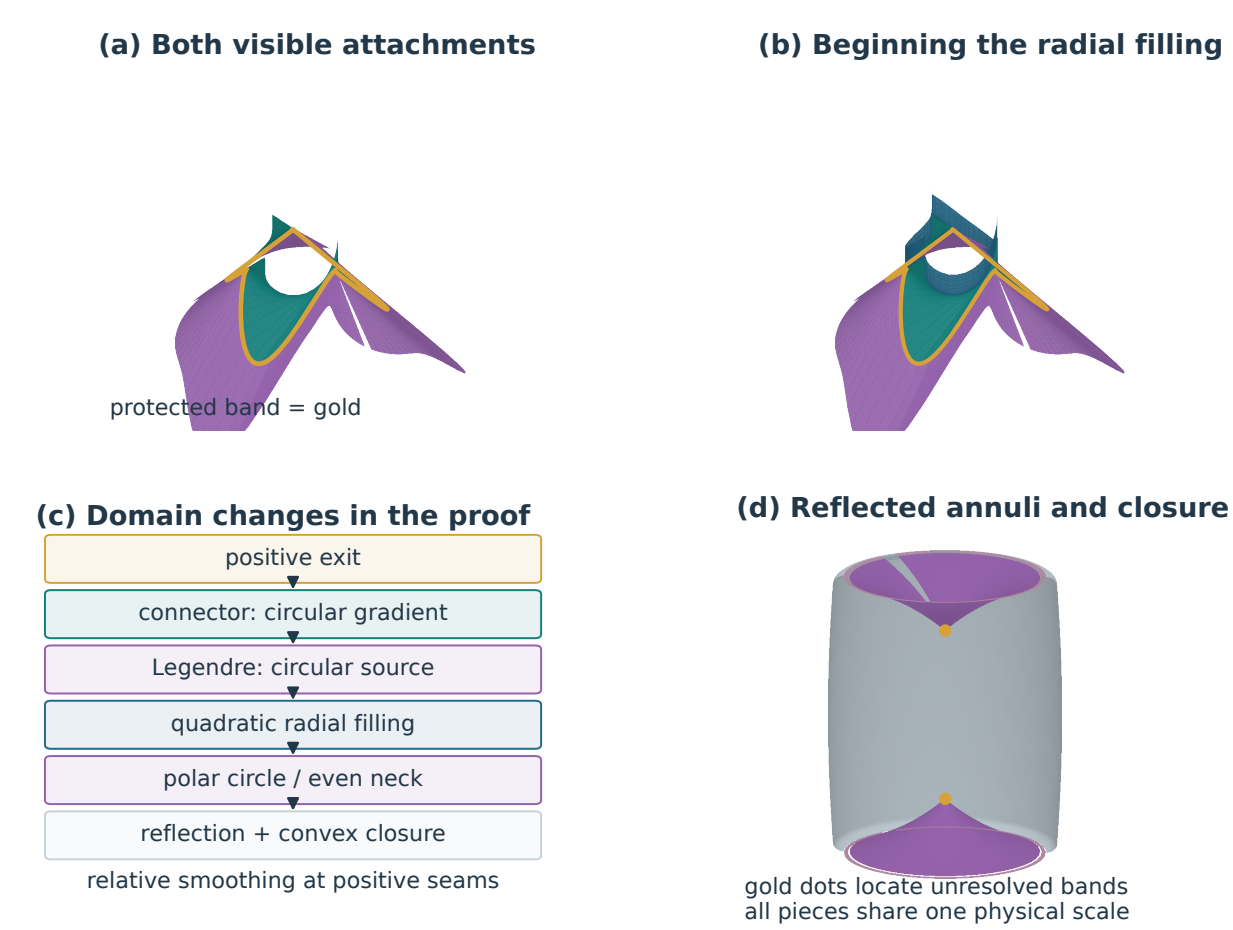


FIGURE 5. Completion of the saved balanced numerical band and the proof's domain changes. (a) The unchanged protected band is gold; the initial outer and inner attachments are teal and purple. (b) The blue patch begins the radial filling beyond the outer connector. Both windows have the same physical bounds and scale. (c) A positive exit feeds a visible connector with circular outgoing gradient. Legendre duality turns that trace into a circular source seam for quadratic radial filling. The two ends then receive the polar and neck profiles described in Proposition 7.5. (d) A cutaway of the northern assembly, its reflection, and the gray convex exterior, all under one common similarity. Gold location markers are enlarged for visibility: the band is approximately 1.17×10^{-12} of the total height in this saved assembly. The mesh matches first jets at internal interfaces and is shown before relative smoothing; it does not certify a smooth tight torus. Lemma 7.1 and Section 8 supply the smooth geometric completion in the proof.

8. From the radial support to a tight torus

Proposition 7.5 gives two explicit end germs without passing through a logarithmic support. One end resolves into a circular boundary, the other into a reflectable parabolic neck. We then close the resulting negative-curvature annulus by a convex annulus. This stage uses only the support geometry. It transports the protected bending region by one fixed ambient isometry.

8.1. **Circular compactification and reflection.** Let $\widetilde{G}$ be the completed potential in (7.17), and put $r = |p|$. For ambient points $(v, z) \in \mathbb{R}^2 \times \mathbb{R}$, write $\rho_{\text{spatial}} = |v|$. Using (2.3), its image near the puncture is

$$X_{\widetilde{G}}(r, \theta) = ((R_N - \mu r)e_\theta,\ d_0 + \tfrac{\mu}{2} r^2). \tag{8.1}$$

Near infinity it is

$$X_{\widetilde{G}}(r, \theta) = ((A + B/r^2)e_\theta,\ d_\infty - 2B/r). \tag{8.2}$$

Apply the ambient isometry $T(x) = -x + d_\infty e_3$ to the whole immersion, and shift the angular parameter by π. We use this same normalization for the protected core and its bending fields: if their original restrictions are $X|_V$ and $y|_V$, their restrictions after normalization are $T \circ X|_V$ and $dT(y)|_V = -y|_V$. Thus subsequent relative statements are literal in the normalized coordinates. At the puncture, write $t = -r$; then the northern boundary collar is

$$F_N(\theta, t) = (R_N + \mu t)e_\theta + \big(h - \tfrac{\mu}{2} t^2\big)e_3, \qquad -\delta < t \le 0, \tag{8.3}$$

where $h = d_\infty - d_0$. At the other end put $\eta = 1/r$. The neck is

$$X(\theta, \eta) = (A + B\eta^2)e_\theta + 2B\eta e_3, \qquad 0 \le \eta < \eta_0, \tag{8.4}$$

for sufficiently small $\eta_0 > 0$. Its radius and height satisfy

$$\rho_{\text{spatial}} = A + \frac{z^2}{4B}. \tag{8.5}$$

It is therefore smoothly reflected across $z = 0$.

The puncture $p = 0$ is *not* filled by one surface point: it resolves into a whole boundary circle. In (8.3), the angular and transverse derivatives at $t = 0$ are $R_N J e_\theta$ and μe_θ, so the immersion is regular. The neck (8.4) is also regular at $\eta = 0$, where the corresponding derivatives are AJe_θ and $2Be_3$.

Proposition 8.1 (Completed saddle annulus)**.** *The completed support compactifies, after reflection at its neck, to a smooth embedded annulus N with $K < 0$ in its interior. Its boundary circles have common radius R_N, at heights $\pm h$ with $h > 0$. Near them its collars are* (8.3) *and*

$$F_S(\theta, t) = (R_N + \mu t)e_\theta + \big(-h + \tfrac{\mu}{2} t^2\big)e_3, \qquad -\delta < t \le 0. \tag{8.6}$$

On either collar,

$$K = \frac{t}{\mu(R_N + \mu t)(1 + t^2)^2}. \tag{8.7}$$

All bending fields supported in the protected region extend by zero over the annulus, and vanish on full boundary collars.

Proof. Embeddedness of the upper half. By (7.18) its horizontal projection is globally injective on the punctured source. The two displayed end germs show that projection extends to nested disjoint boundary circles of radii $A < R_N$, with regular full surface collars. Consequently the compactified upper half is embedded.

Height separation. The interior Gauss normal is represented by $q(p) = (p, 1)/\sqrt{1 + |p|^2}$, up to orientation, and is never vertical for $0 < |p| < \infty$. Hence the vertical height has no interior critical point. Formula (8.4) gives strictly positive height close to the lower neck circle, while (8.3) gives

height strictly below h near the upper circle. The maximum and minimum on the compact annulus therefore occur at these boundary components. It follows that

$$h > 0, \qquad 0 < X_3 < h \quad \text{in the upper interior.}$$

The reflection at the even neck (8.5) is smooth; the two halves have disjoint interiors, so their union is embedded and has negative curvature throughout the interior.

Boundary curvature and retained fields. For (8.3), set $r(t) = R_N + \mu t$ and $z(t) = h - \mu t^2/2$. A unit normal is $(te_\theta + e_3)/\sqrt{1+t^2}$; the metric coefficients are $r(t)^2$ and $\mu^2(1+t^2)$, and the second fundamental coefficients are $-r(t)t/\sqrt{1+t^2}$ and $-\mu/\sqrt{1+t^2}$. This proves (8.7). Reflection gives the same intrinsic formula at the southern end. Every change was outside the protected region, where all retained bendings vanish on a collar; extension by zero is therefore smooth and preserves their linearized metric equations. □

8.2. **Convex closure.** The two boundary circles of the saddle annulus have radius R_N and heights $\pm h$. We join them by a positive-curvature annulus lying outside the cylinder of radius R_N. A concave radius as a function of height gives a convex lateral surface. At each end we prescribe the entire parabolic germ, so the final join is smooth.

Lemma 8.2 (Convex closure)**.** *For any $R_N, \mu, h > 0$, there exists a convex body whose boundary consists of the disks*

$$D_\pm = \{(v, \pm h) : |v| \le R_N\}$$

and a smooth annulus U_{sp} with strictly positive curvature in its interior. The annulus agrees with (8.3) *and* (8.6) *for $0 \le t < \delta$, has outward Gauss map*

$$\nu : U_{\mathrm{sp}}^\circ \longrightarrow \mathbb{S}^2 \setminus \{\pm e_3\} \tag{8.8}$$

a diffeomorphism, and satisfies

$$U_{\mathrm{sp}}^\circ \subset \{\rho_{\mathrm{spatial}} > R_N,\ |z| < h\}, \qquad U_{\mathrm{sp}} \neq -U_{\mathrm{sp}}. \tag{8.9}$$

Proof. Choose an even function r_0, continuous on $[-h, h]$ and smooth with $r_0'' < 0$ on $(-h, h)$, with $r_0(\pm h) = R_N$, $r_0 > R_N$ in the interior, and near the endpoints

$$r_0(z) = R_N + \sqrt{2\mu(h - z)} \quad (z \text{ near } h), \qquad r_0(z) = R_N + \sqrt{2\mu(h + z)} \quad (z \text{ near } -h). \tag{8.10}$$

For completeness, on $[0, h)$ choose a smooth function $v = -r_0'$ with $v' > 0$, $v(z) = \epsilon z$ near zero, and $v(z) = \sqrt{\mu/[2(h-z)]}$ near h. Choose the endpoint neighborhoods small enough to interpolate with $v' > 0$ between these germs, and set $r_0(z) = R_N + \int_z^h v(s)\, ds$ for $z < h$, with $r_0(h) = R_N$. The endpoint singularity of v is integrable. The expression is smooth after even reflection at zero, and its lateral surface is smooth at the endpoints in the t coordinate used below. In the height coordinate, r_0 is continuous on $[-h, h]$ and smooth on $(-h, h)$; its endpoint slopes are infinite. It is strictly concave and agrees with (8.10) at the upper endpoint.

Choose a nonzero smooth $b \ge 0$ supported in $(0, h)$ and set $r = r_0 + \lambda b$. For sufficiently small $\lambda > 0$ it remains strictly concave, has unchanged end germs, and is no longer even. The body $\mathcal{K} = \{(v, z) : |z| \le h,\ |v| \le r(z)\}$ is convex, since r is concave. Its lateral annulus $(r(z)e_\theta, z)$ has

$$\nu = \frac{(e_\theta, -r'(z))}{\sqrt{1 + r'(z)^2}}, \qquad K = -\frac{r''(z)}{r(z)(1 + r'(z)^2)^2} > 0.$$

As z increases, r' decreases from $+\infty$ to $-\infty$, which gives the stated Gauss diffeomorphism. The upper equality in (8.10), together with $z = h - \mu t^2/2$, gives $r = R_N + \mu t$ for $t \ge 0$; the southern equality works identically. Thus the lateral annulus continues both parabolic collar formulas smoothly across $t = 0$. Since $r(z) > R_N$ for $|z| < h$, the lateral annulus lies outside the prescribed cylinder, and non-evenness excludes central symmetry. The disks close the auxiliary convex body.

Smoothness of its boundary across the disk rims is not needed: the smooth torus is obtained by joining the lateral annulus to the saddle collars. □

The global gradient image in (7.18) gives $\rho_{\text{spatial}} < R_N$ on N°, and the height separation in Proposition 8.1 gives $|z| < h$ there. Hence

$$N \subset \mathcal{C} := \{\rho_{\text{spatial}} \le R_N,\ |z| \le h\},$$

with strict inclusion for interior points. By (8.9), U_{sp} meets N only along their two boundary circles. On the two sides, their collars are restrictions of the same formulas (8.3) and (8.6) to $t \le 0$ and $t \ge 0$, respectively. They form a smooth embedded torus. The curvature signs are negative on N°, positive on U_{sp}°, and zero exactly on the two joining circles. By the Gauss area formula and (8.8),

$$\int_{U_{\text{sp}}^\circ} K\, dA = 4\pi. \tag{8.11}$$

Gauss–Bonnet gives total absolute curvature 8π, hence tightness. This uses the standard tightness characterization in terms of total absolute curvature [10].

8.3. **An asymmetric fixed region.** For a set $A \subset \mathbb{R}^3$, $\operatorname{Stab}_{\text{Isom}(\mathbb{R}^3)} A$ is the group of Euclidean isometries carrying A onto itself. We call an embedding $X : M \to \mathbb{R}^3$ *marked* when

$$\operatorname{Stab}_{\text{Isom}(\mathbb{R}^3)} X(\{K_X > 0\}) = \{\text{Id}\}.$$

Any ambient congruence between embeddings with this same positive-curvature image must therefore be the identity.

Lemma 8.3 (Small affine marking)**.** *For every sufficiently small $\epsilon > 0$, the map*

$$A_\epsilon = \begin{pmatrix} 1 & 0 & \epsilon \\ 0 & 1+\epsilon & 2\epsilon \\ 0 & 0 & 1 \end{pmatrix}$$

tends to Id *and satisfies* $\operatorname{Stab}_{\text{Isom}(\mathbb{R}^3)} A_\epsilon(U_{\text{sp}}^\circ) = \{\text{Id}\}$.

Proof. An isometry preserving the open annulus preserves its closure and permutes its two boundary ellipses. Their centers are $\pm h(\epsilon, 2\epsilon, 1)$; their planes are horizontal; their unequal semiaxes lie along e_1, e_2 and have lengths $R_N, (1+\epsilon)R_N$. The midpoint of the centers is fixed, so the translation is zero. The planes and principal axes force the orthogonal part to be diagonal with entries ± 1. If both centers are fixed, all entries are $+1$, because each coordinate of $(\epsilon, 2\epsilon, 1)$ is nonzero. If they are exchanged, all are -1. This would give central inversion of the annulus and hence $U_{\text{sp}} = -U_{\text{sp}}$, contrary to Lemma 8.2. Only the identity remains. □

8.4. **The completed protected bending family.**

Theorem 8.4 (Protected identity-return realization)**.** *There exist a smooth torus M, a tight embedding $X : M \to \mathbb{R}^3$, a compact annulus $V \Subset \{K_X < 0\}$, an open annular neighborhood $W \supset V$, and a nonempty open interval I such that:*

(i) *The Gauss map and horizontal projection are injective on W. In planar support coordinates,* $\det D^2 G < 0$ *and both the source and gradient maps are diffeomorphisms onto planar annuli.*

(ii) *Under the fixed source identification with the protected ruled band, the transported field $\bar{y}_F = -y_F$, with y_F given by* (3.5)*, defines an injective linear map from $C_c^\infty(I)$ to bendings supported in V. It is extended by zero outside that band. Disjoint profile supports give disjoint bending supports.*

(iii) *Once the balanced speed is fixed and the core has undergone the fixed ambient normalization of Section 8, the completion leaves a prescribed compact subannulus unchanged. Every sufficiently small L^1 speed change preserving closure, period balance, and visibility admits the same type of completion, with its own protected core.*

(iv) *The embedding has no open planar subset and total absolute curvature 8π.*

Proof. Build and protect the local bending family. Lemma 5.2 and Theorem 5.5 give a balanced embedded frame with visible traces. Equation (5.16) gives $I_0 = 0$, so Theorem 3.3 provides the kernel. Shrink the band for Gauss and projection injectivity, and choose the protected level annulus V_0. Enlarge it slightly to a compact V, still separated from both exit collars. All fields from $C_c^\infty(I)$ are supported there.

Complete outside the protected neighborhood. The central mixed-entry formula and exit lemma give two positive seams. The connector and quadratic radial filling lemma, combined with Legendre duality and the neck adapter, give the two-sided support of Proposition 7.5. Proposition 8.1 yields the embedded saddle annulus, and Lemma 8.2 closes it into an embedded torus. Apart from the one ambient normalization applied uniformly to the core and fields, every change is outside a neighborhood of V, so the transported bendings extend smoothly by zero. The positive curvature integral is 4π; Gauss–Bonnet gives 8π absolute curvature and tightness. Curvature vanishes only on two circles, so no planar open set exists.

Record the relative scope of the construction. Small L^1 speed changes give small positional changes with tangent directions fixed. Embedding and visibility persist as in Theorem 5.5. For each final balanced speed we choose its thin band and exit collars and repeat the completion, leaving its prescribed normalized core fixed. This asserts existence of the same type of completion, not an identification of different cores or a common exterior for all speeds. □

Corollary 8.5 (Marked protected bending family)**.** *There are a tight torus embedding $X : M \to \mathbb{R}^3$, a compact $V \Subset \{K_X < 0\}$, a nonempty open interval I, and an injective linear map $\mathcal{Y} : C_c^\infty(I) \to C^\infty(M, \mathbb{R}^3)$ whose values are bendings supported in V, preserving disjointness of supports, such that*

$$\mathrm{Stab}_{\mathrm{Isom}(\mathbb{R}^3)} X(\{K_X > 0\}) = \{\mathrm{Id}\}.$$

The torus has no open planar subset and total absolute curvature 8π.

Proof. Apply A_ϵ to the preceding torus and transport fields by (2.6). On the protected band the resulting map is $\mathcal{Y}(F) = A_\epsilon^{-T}\bar{y}_F = -A_\epsilon^{-T} y_F$, using the same source identification, and is extended by zero. This preserves embedding, curvature signs, tightness, strain, and supports. Lemma 8.3 marks the positive-curvature region. The equality case of (2.1) gives the same absolute curvature value after the affine change. □

9. Exact metric branching

Fix the marked tight torus from Corollary 8.5. Its reference metric is g_X. The sign branches constructed in this section induce a new common metric g, given by a nonnegative quadratic correction to g_X. We first make one exact pair, then choose finitely or countably many independent signs. Tightness follows from the unchanged positive-curvature region, and image noncongruence follows from its marking together with intrinsic isometry rigidity.

9.1. **Finite independent signs.** The metric reconstruction map has the exact quadratic expansion

$$\mathcal{G}(X + y) = \mathcal{G}(X) + S_X y + \langle dy, dy\rangle. \tag{9.1}$$

Lemma 9.1 (Disjoint quadratic branching)**.** *If $y_1, \dots, y_m \in \ker S_X$ have pairwise disjoint supports, then*

$$g_{X+\sum_{j=1}^m \varepsilon_j a_j y_j} = g_X + \sum_{j=1}^m a_j^2 \langle dy_j, dy_j\rangle, \qquad \varepsilon_j \in \{-1, 1\}. \tag{9.2}$$

The same identity holds for an infinite series converging in C^∞.

Proof. Every linear term vanishes by the strain equation. A derivative is zero off its field's support, so disjoint supports kill every mixed quadratic term pointwise. This proves the finite formula. Smooth convergence allows passage to the limit in the metric and its derivatives. □

Two separate issues remain: tightness of the perturbed embeddings, and noncongruence of their images. Both are controlled by the unchanged region.

Proposition 9.2 (Relative preservation of absolute curvature)**.** *Let $X : M \to \mathbb{R}^3$ be an embedding of a closed surface and let $V \Subset \{K_X < 0\}$. Every sufficiently small C^2 perturbation supported in V remains an embedding and has the same total absolute curvature. If X is tight, so is the perturbation.*

Proof. Compactness and smallness preserve negative curvature on V and preserve embedding. Off V the map is unchanged, so the positive-curvature region P and its curvature integral are unchanged. For either surface, Gauss–Bonnet gives

$$\int_M |K|\, dA = 2\int_P K\, dA - 2\pi\chi(M).$$

Thus the absolute curvature is unchanged, and equality in (2.1) preserves tightness. □

Lemma 9.3 (An intrinsic isometry fixed on an open set)**.** *An isometry of a connected smooth Riemannian manifold which fixes a nonempty open set pointwise is the identity. The same holds for a connected manifold with boundary when the fixed set meets its interior.*

Proof. In the interior, the set of points fixed with identity differential is nonempty and closed. It is open by the exponential-map relation $f(\exp_p v) = \exp_{f(p)}(df_p v)$. Connectedness propagates the identity. For a manifold with boundary, apply this to its connected interior and then use continuity at the boundary. □

Proposition 9.4 (Finite sign families)**.** *Let $X : M \to \mathbb{R}^3$ be a tight embedding of a closed connected surface, with $P = \{K_X > 0\}$ and $\mathrm{Stab}_{\mathrm{Isom}(\mathbb{R}^3)} X(P) = \{\mathrm{Id}\}$. Let $y_1, \dots, y_m$ be nonzero bendings with disjoint supports in one compact $V \Subset \{K_X < 0\}$. For all sufficiently small positive a_j, the 2^m maps in* (9.2) *are tight embeddings with a common metric, agree off V, and have pairwise noncongruent images.*

Proof. The quadratic lemma gives the common metric and the curvature proposition gives tight embeddings. Suppose an ambient isometry carries one image onto another. Gaussian curvature is preserved, so it must carry the unchanged positive region $X(P)$ onto itself. The stabilizer hypothesis makes that ambient isometry the identity.

The images therefore coincide. If their parametrizations are X_ε and X_δ, then $\varphi = X_\delta^{-1} \circ X_\varepsilon$ is an isometry of the common intrinsic metric. It fixes P pointwise because both maps equal X there. The fixed-open-set lemma gives $\varphi = \mathrm{Id}$. Thus the maps coincide pointwise. On a point where $y_j \neq 0$, all other fields vanish, so $\varepsilon_j = \delta_j$. This holds for every j. □

For $m = 1$, this is already the exact nonuniqueness assertion. Increasing m repeats the same localized mechanism independently.

Figure 6 sketches the passage from independent local supports to the infinite sign space. Disjoint intervals in the leaf coordinate give disjoint bending supports on the surface. The metric sees the square of each amplitude and loses its sign. Rapid decay in every derivative order makes the infinite series smooth, including at an accumulation of the supports. The fixed marked region then separates the resulting congruence classes.

9.2. **Countably many signs and the quotient topology.**

Lemma 9.5 (Hausdorff Euclidean orbit space)**.** *For a closed Riemannian manifold (M, g), $\mathrm{Emb}_g(M, \mathbb{R}^3)/\mathrm{Isom}(\mathbb{R}^3)$ is metrizable and hence Hausdorff.*

Proof. Subtract the mean of each embedding with respect to the Riemannian volume $d\mu_g$. Centering removes translations, so the quotient is the quotient of centered embeddings by $O(3)$. On that space use the invariant metric

$$d(f, h) = \sum_{k \geq 0} 2^{-k-1} \min\{1, \|f - h\|_{C^k(g)}\}.$$

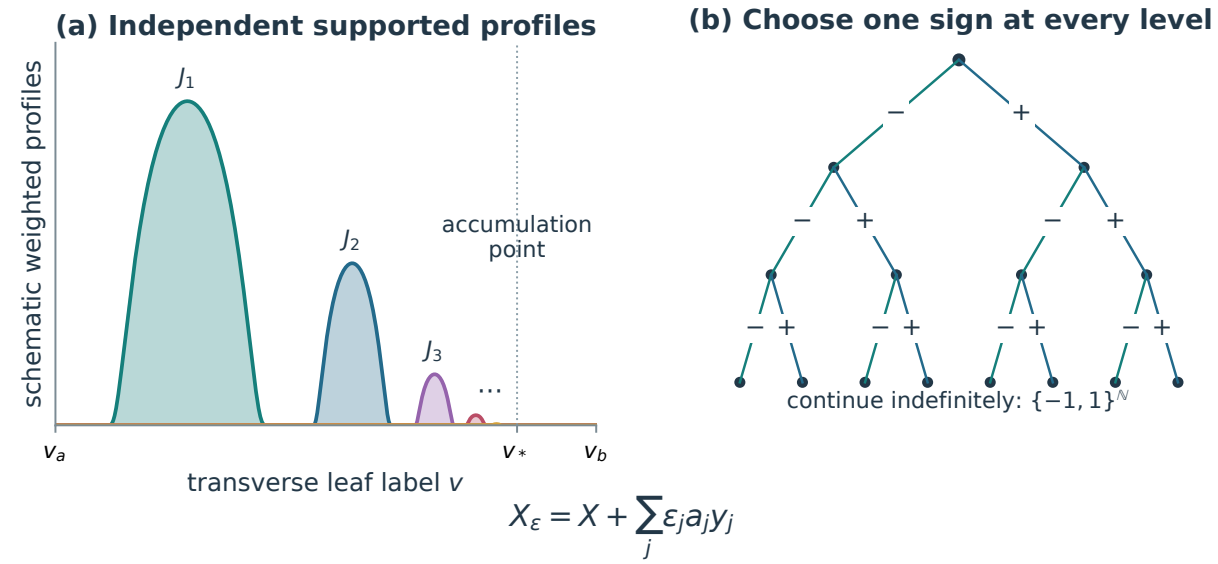


FIGURE 6. The Cantor family as independent sign choices. (a) Schematic weighted test profiles on disjoint transverse intervals $J_j \Subset I$ accumulating at an interior leaf label. The associated fields $y_j = \mathcal{Y}F_j$ have supports lying in disjoint annuli. The plotted heights only illustrate decay; the proof chooses amplitudes controlling all required C^k norms. (b) Three finite levels show 2, 4, and 8 choices; continuing gives $\{-1,1\}^{\mathbb{N}}$ with the product topology. Disjoint supports eliminate mixed metric terms, and therefore every branch has the same metric $g_X + \sum_j a_j^2 \langle dy_j, dy_j\rangle$. Theorem 9.6 proves smooth convergence, tightness, noncongruence, and the embedding into the metric fiber modulo Euclidean isometries. This totally disconnected parameter family is not a continuous isometric flex.

Here $\|f\|_{C^k(g)} = \max_{0\le j\le k} \sup_M |\nabla^j f|$, using the Levi–Civita connection of g and the Euclidean target norm. Since $O(3)$ is compact, the orbit distance $\bar{d}([f],[h]) = \min_{Q\in O(3)} d(f, Qh)$ is a metric. Invariant metric balls show that it induces the quotient topology. □

Theorem 9.6 (Localized sign families on a marked tight surface)**.** *Let $X : M \to \mathbb{R}^3$ be a tight embedding of a closed connected surface whose positive-curvature image has the trivial ambient stabilizer required in Proposition 9.4. Let y_j, $j \ge 1$, be nonzero bendings with pairwise disjoint supports in one compact $V \Subset \{K_X < 0\}$. Choose positive amplitudes so that the perturbation series $\sum_{j\ge1} \varepsilon_j a_j y_j$ converges in every C^k norm uniformly in the signs and its sum is sufficiently small in C^2. Set*

$$X_\varepsilon = X + \sum_{j\ge1} \varepsilon_j a_j y_j, \qquad \varepsilon \in \{-1,1\}^{\mathbb{N}}.$$

Then the maps are tight embeddings with the common smooth metric

$$g = g_X + \sum_{j\ge1} a_j^2 \langle dy_j, dy_j\rangle.$$

Their images are pairwise noncongruent, and the sign map induces a topological embedding into $\mathrm{Emb}_g(M,\mathbb{R}^3)/\mathrm{Isom}(\mathbb{R}^3)$. Such amplitudes always exist.

Proof. Choose a uniformly smooth and small series. For a prescribed small $\epsilon > 0$, require

$$a_j \max_{0\le k\le j+2} \|y_j\|_{C^k} < \epsilon 2^{-j}.$$

For each derivative order the tail is uniformly summable, independently of the signs. The full sum is also uniformly small in C^2. The quadratic identity and the curvature proposition give the common metric and tight embeddings, with the positive region unchanged.

Rule out congruence and reparametrization. The argument in Proposition 9.4 applies verbatim: an ambient congruence preserves $X(P)$ and is the identity; the resulting intrinsic reparametrization fixes P and is the identity. At a point where $y_j \neq 0$, disjointness then determines its sign. Thus every sign sequence gives a different congruence class.

Verify the Cantor topology. Agreement of sufficiently many initial signs makes the difference of two maps a uniformly small tail in each C^k norm. Hence the sign map is continuous for the product topology. Its domain is compact and its quotient target is Hausdorff by Lemma 9.5. A continuous injection from a compact space into a Hausdorff space is a topological embedding. □

10. Consequences of the protected band

10.1. **The tight metric fiber.**

Proof of Theorem 1.1. Use the marked bending family in Corollary 8.5. Choose disjoint open intervals $J_j \Subset I$ and nonzero $F_j \in C_c^\infty(J_j)$, and let $y_j = \mathcal{Y}F_j$. The map $\mathcal{Y}$ preserves disjoint supports. Choose the amplitudes as in Theorem 9.6. It gives a common metric, tight embeddings, noncongruent images, and the Cantor embedding into the quotient. All maps equal X off V, in particular on the same nonempty positive-curvature open set.

The reference curvature vanishes only on the two joining circles. The perturbed curvature remains negative on V and is unchanged elsewhere, so no member has an open planar subset. Tightness gives total absolute curvature 8π. Selecting any two distinct sign sequences proves the first assertion with the same metric. The finite proposition also proves the two-realization phenomenon directly from one nonzero field. □

Corollary 10.1 (Smooth tight nonrigidity)**.** *Smooth embedded tight tori are not determined up to congruence by their intrinsic metrics. One such metric has a Cantor family of noncongruent realizations.*

Proof. Apply Theorem 1.1. □

10.2. **Closed asymptotic curves and zero linking.**

Proof of Theorem 1.2. Take N from Proposition 8.1. The collar formulas give its circular tangent-plane boundaries and negative interior curvature. The northern Gauss coordinates cover the northern hemisphere with its pole removed. Reflection covers the southern hemisphere with its pole removed; the smooth neck maps to the equator. Each half is parametrized by its normal, and the neck has nonzero curvature. Together these give a Gauss diffeomorphism onto the twice-punctured sphere.

The protected leaves in (3.7) remain embedded and asymptotic. To compute their normal linking, cap N by the planar disks $D_\pm$ at heights $\pm h$. The height bounds show that $S = N \cup D_+ \cup D_-$ is an embedded topological sphere. A small normal push-off of a protected leaf is disjoint from S: near the leaf this follows from a tubular neighborhood, and away from the leaf from compact separation. The leaf bounds a disk in S disjoint from the push-off. Intersection with that disk computes linking zero. □

Corollary 10.2 (Degenerate asymptotic return on a tight torus)**.** *A smooth embedded tight torus contains an open annulus of closed asymptotic curves in its negative-curvature region such that every*

leaf satisfies

$$\int_\Gamma k_g k_n |K|^{-1/2}\, ds = 0.$$

Here $k_g = \langle \nabla_T T, Z\rangle$ *and* $k_n = \mathrm{II}(Z,Z)$*, with* Z *a unit tangent perpendicular to the oriented leaf tangent* T*.*

Proof. Use the unmarked torus in Theorem 8.4. Its protected return is the identity, so each leaf has multiplier one. Lemma 3.4 gives the integral's vanishing. □

Corollary 10.3 (Fixed-boundary isometric saddle annuli)**.** *There are two noncongruent smooth embedded annuli with strictly negative Gaussian curvature, the same induced metric, and the same two convex planar boundary curves. They agree on full boundary collars.*

Proof. Choose strictly negative boundaries and eliminate their symmetries. Cut the northern and southern collars of N at $t = -\eta_+$ and $t = -\eta_-$, where $0 < \eta_+ \neq \eta_- \ll 1$. Retain the protected support. These cuts give strictly negative curvature even at the boundary and distinct circular radii. Apply A_ϵ from Lemma 8.3 and transport the bendings by (2.6). The new boundary curves are noncircular ellipses in parallel planes with different semiaxes, so an isometry preserving their union must preserve each individually.

Their principal axes force the orthogonal part of such an isometry to be diagonal. Their centers are distinct multiples of $(\epsilon, 2\epsilon, 1)$. Fixing both centers forces all three diagonal signs to be $+1$, and then the translation is zero. Thus their union has trivial ambient stabilizer.

Construct the exact pair and exclude identical images. Let $\widetilde{X}$ be the truncated affine embedding and $y \neq 0$ a retained bending. For small $t > 0$, $\widetilde{X} \pm ty$ are embedded strictly negative-curvature annuli. They agree on full boundary collars, and (9.1) gives a common metric. Any ambient congruence must preserve the boundary union and is therefore the identity. If the images coincided, their induced reparametrization would be an intrinsic isometry fixing the common collar pointwise. Lemma 9.3 would make it the identity on the whole annulus, forcing $y = 0$. This contradiction proves noncongruence. □

10.3. **Complete negatively curved ends.** For a complete cylinder we cut the saddle annulus before its zero-curvature boundary circles and attach strictly convex radial profiles. Agreement of full germs at the cuts preserves smoothness and the compact bending family.

Lemma 10.4 (Strictly convex continuation of a radial profile)**.** *Suppose* q_0 *is smooth near* H*, with* $q_0(H) > 0$*,* $q_0'(H) > 0$*, and* $q_0'' > 0$ *there. For every* $\beta > 0$ *there is a smooth* q *on* $[H, \infty)$ *equal to* q_0 *near* H*, with* $q, q', q'' > 0$*, and with* $q'' = \beta$ *eventually. In particular* $q'(z) \to \infty$*.*

Proof. Choose $d > 0$ with $q_0'' > 0$ on $[H, H + 2d]$. A smooth cutoff from zero near H to one for $z \geq H + d$ gives a positive smooth function b equal to q_0'' near H and to β for $z \geq H + d$. Define

$$q(z) = q_0(H) + q_0'(H)(z - H) + \int_H^z (z - v) b(v)\, dv. \tag{10.1}$$

Then $q'' = b > 0$, so q' and q stay positive. Near H the second derivative and initial first jet agree with those of q_0, giving full agreement there. Eventually q is quadratic with positive second derivative. □

Proof of Theorem 1.3. Step 1: cut at strictly negatively curved circles. Use the unmarked N of Proposition 8.1 and choose sufficiently small $0 < \eta < \delta$ so that cutting both collars at $t = -\eta$ retains the protected support and $H > 0$. Put

$$R_\eta = R_N - \mu\eta > 0, \qquad H = h - \frac{\mu}{2}\eta^2 > 0. \tag{10.2}$$

The retained annulus N_η has boundary circles at heights $\pm H$ and radius R_η, with $K < 0$ everywhere up to those boundaries. Its interior lies between these heights because the height function has no interior critical point.

Step 2: extend the circular germs to convex ends. The northern collar, solved for radius as a function of height, is

$$\begin{aligned} q_0(z) &= R_N - \sqrt{2\mu(h-z)}, \\ q_0'(z) &= \frac{\mu}{\sqrt{2\mu(h-z)}}, \qquad q_0''(z) = \frac{\mu^2}{[2\mu(h-z)]^{3/2}}. \end{aligned} \tag{10.3}$$

At $z = H$ one has $q_0 = R_\eta$, $q_0' = 1/\eta > 0$, and $q_0'' = 1/(\mu\eta^3) > 0$. The continuation lemma gives q with the same germ and eventually quadratic growth. Attach

$$E_\pm(\theta, z) = q(z)e_\theta \pm z e_3, \qquad z \in [H, \infty). \tag{10.4}$$

Full-germ agreement makes both joins smooth. The resulting source is $S^1 \times \mathbb{R}$ with no boundary.

Step 3: verify curvature, embedding, and completeness. For the upper end,

$$g = q^2 d\theta^2 + (1 + (q')^2)dz^2, \qquad n_+ = \frac{-e_\theta + q' e_3}{\sqrt{1 + (q')^2}}. \tag{10.5}$$

Its second fundamental coefficients are $q/\sqrt{1 + (q')^2}$ and $-q''/\sqrt{1 + (q')^2}$, with zero mixed coefficient, so

$$K = -\frac{q''}{q(1 + (q')^2)^2} < 0. \tag{10.6}$$

Reflection gives the same result below, including at both smooth seams. Height determines z on each end and its positive radius determines θ, so each end is embedded. Their interiors lie strictly above or below the core's height range, excluding all intersections.

The embedding is proper: a compact ambient set bounds height and hence bounds the end parameters, leaving only a compact part of the source. An intrinsic Cauchy sequence has bounded Euclidean image, since induced distance dominates ambient distance. Properness gives a subsequence converging on the surface, and local smooth coordinates turn this into intrinsic convergence. The Cauchy sequence therefore converges, proving completeness.

Step 4: identify the Gauss map and curvature integral. At the upper join $n_+ = (-\eta e_\theta + e_3)/\sqrt{1 + \eta^2}$, matching the core normal. Along the end,

$$\frac{d}{dz}\frac{q'}{\sqrt{1 + (q')^2}} = \frac{q''}{(1 + (q')^2)^{3/2}} > 0, \qquad \frac{q'}{\sqrt{1 + (q')^2}} \longrightarrow 1.$$

Writing $\zeta_\eta = (1 + \eta^2)^{-1/2}$, the upper end covers the punctured cap $\zeta_\eta \le n_3 < 1$ once, and reflection covers $-1 < n_3 \le -\zeta_\eta$ below. The retained core covers $|n_3| \le \zeta_\eta$ once by Theorem 1.2. The descriptions agree at the seams, where $K \neq 0$, so the global normal is a diffeomorphism onto the twice-punctured sphere. Its area gives $\int_\Sigma |K|\, dA = 4\pi$. For comparison, each end contributes

$$2\pi \int_H^\infty \frac{q''}{(1 + (q')^2)^{3/2}}\, dz = 2\pi(1 - \zeta_\eta).$$

Step 5: retain the closed leaves and bending family. The surface is unchanged near the protected band, so its closed leaves, identity return, and normal fields along the leaves are unchanged. Their linking remains zero. All bendings vanish near the cuts and extend by zero to the ends, retaining their strain equations, injectivity, and common compact support. □

The eventual quadratic profiles have $K \to 0$ at infinity. Completeness therefore comes with strict negativity everywhere, but no uniform negative upper bound. The result addresses the

embedded-existence question in [9, after Theorem 3.4]; their plane-topology obstruction does not apply to a cylinder.

Acknowledgements

I thank my mentor Xiang Ma for communicating with me about this passage.

AI usage and my personal words

This paper was written with "assistance" from (in fact, entirely by) ChatGPT-5.6-Sol, ChatGPT-6-Astra, and ChatGPT-6.1-Sol.

As of October 10, 2026, I now claim the correctness of the paper because the most important theorem compiles without `sorry` or extra axioms. See the repository for details. The remaining risks are:

(1) I have only falsified the rigidity conjecture; I have not done the other propositions, such as the Cantor family.
(2) The translation of the theorem might not be entirely risk-free. AI has checked it, and it is highly unlikely to mislead us, but one can never be certain.
(3) Mistranslations might occur during the process, and the manuscript's lemmas and exact proof details might not be entirely correct. The formalization guarantees the existence of the band and the counterexample, but the other steps might not be accurately represented. The manuscript will undergo many edits after the formalization, and these edits might deviate further from the formalization's route. In fact, this faces the same problem as the Navier–Stokes equation proof. Yet, if we treat natural-language manuscripts as sketches and roadmaps, guiding a correct Lean proof already guarantees their trustworthiness. You can always ask an AI to fill a gap if you find one; having a correct roadmap gives you confidence that a way to fill it can be found.
(4) The Lean kernel might contain bugs. I do not think this is likely to be a problem, and if there really is a problem, it is hardly mine alone. If the sky falls, the tall guys will hold it up.

I had not noticed this problem until Levent Alpoge produced a counterexample to the Carathéodory conjecture, followed almost immediately by YTD and S^6. At that point, I began to believe that AI could solve problems in differential geometry. I began asking GPT which problem on Ghomi's list of open problems could be next (Carathéodory was on the list). GPT suggested this one because the analytic case is true, the polyhedral case is false, and the smooth case is unsolved. This resembled the Carathéodory conjecture, where the analytic case is also true but the smooth case was ultimately shown to be false. I then asked chat mode to solve the problem. It found a counterexample to Problem 1.1 in Ghomi's 2025 paper, showing that the simple route toward proving rigidity was false. It did not give any other promising result. Several days later, I decided to make an all-out effort and downloaded Danus, running it for 28 hours and using about 140% of the weekly allowance on the $200 Pro plan. Then Astra came out, and it took two hours and 25% of the Pro allowance to find the counterexample. Although the counterexample has changed considerably, its idea has survived into the final result. Evidence of this timeline can be found in the timestamp archive.

I was shocked, so I asked it to write out the result and considered giving it to my mentor or publishing it quickly. Yet the paper was not a clean formula like the Carathéodory result: it was over 300 pages long and practically impossible to read. The main construction and proof involved PDEs, and I had not even taken a course in PDEs as an undergraduate. For the next several weeks, I worked continuously to make the paper simpler, more readable, and more rigorous:

- Around this time, I also learned that the mathematics community advocates building mathematical theories and regards open problems as golden geese. I therefore tried asking AI to "generalize the theorems introduced into lemmas and 'theories'." The point was to make AI find the "essence" of the new theorems, simplify and generalize their conditions, and identify where they share the same proof. This could have four outcomes: the theories might actually be useful;

the paper might become more reusable because the "theories" would remain even if part of the construction failed; the theorems might become more trustworthy because fewer conditions could reduce the cognitive burden on both AI and humans; and, if AI actually built theories from scratch, it might think more logically and reduce errors. Yet this approach essentially failed. Most of the construction was too specific, and the work did not produce useful theories. It also expanded the paper to 400 pages.

- I read a story online about someone who made substantial progress and had various AI systems check it, but the AI overlooked a fatal gap, leaving a month's effort in vain. This made me think that I might need to formalize my paper. Yet I did not have enough tokens, and the infrastructure for differential geometry was inadequate. The AI made no progress with the formalization. This was the situation in September. After OpenAI released the new infrastructure, I succeeded.
- The final successful route was consulting ChatGPT Pro in chat mode. This led to significant simplification, rewriting, and checking of the paper. First, I asked it to rewrite the paper to remove the branches, which brought it to 200 pages. Then I asked it to brainstorm simplifications, which brought it to 100 pages. Next, I asked it to brainstorm more intensively and find a good route, and then let it choose one; this brought the paper to 16 pages. Finally, I asked it to fill in the gaps, which brought it to around 30 pages. Its length has not changed much since then.

For now, I do not know how I should take credit. In fact, I did not do anything—or perhaps I have become one of those black-hearted mentors exposed online who suppress their students and take all the credit. I am an undergraduate who could not even understand the paper's basic tools, yet buying a $200 GPT subscription gave me a paper that solved a long-standing open problem, one I might never have been able to reach in a lifetime without AI. Previously, this result might have earned me a paper in JDG, but now I do not even think anyone cares: more important problems are being solved every day, and by the time I wrote this, OpenAI had released over 700 results more important than this one.

Yet, as an undergraduate without an original paper (and perhaps without one ever, since we do not know what AI will do next year), I care about receiving credit for at least doing something. I also care about the effort I put into this paper, although it was not much. This feels ironic because my dilemma somehow has the same structure as mathematics itself. For the reasons above, every day that I delay publishing the paper is another day when someone else might obtain the result with a stronger model, leaving my month's effort in vain. Other researchers may also be exploring this problem with AI. Even without my involvement, they might reach and publish the same result within a month, or even within days. On the other hand, any mathematician who has spent years working seriously on this problem might feel that their efforts have been overtaken by the publication of this result.

The accompanying GitHub repository contains the manuscript sources and figures, interactive numerical visualizations, Lean sources and formalization reports, and the timestamp archive:
`https://github.com/zhangyuetong/smooth-tight-nonrigidity`

Appendix A. General ruled bands and projective return

This appendix generalizes the principal-normal theory. It is not needed for the explicit core or its completion. The general return has two projective parameters rather than the single period I_0, and the bending equation transports a transverse vector field rather than a scalar alone.

Proposition A.1 (Riccati return and a two-jet criterion)**.** *Let $L > 0$ and $X(t,u) = c(t) + ue(t)$ on $(\mathbb{R}/L\mathbb{Z}) \times (-b, b)$, with $|e| = 1$, smooth L-periodic c, e, and $\Delta = \det(c', e, e') \neq 0$. Suppose $u = 0$ is a closed nonruling asymptotic leaf. Then the nonruling characteristics satisfy*

$$u' = q(t,u) = q_1(t)u + q_2(t)u^2, \qquad q = -\frac{\langle c'' + ue'', (c' + ue') \times e \rangle}{2\Delta}. \tag{A.1}$$

With

$$A(t) = \int_0^t q_1(s)\,ds, \quad \mu = e^{A(L)}, \quad \nu = \int_0^L q_2(t)e^{A(t)}\,dt,$$

the local return is

$$\mathcal{R}(u) = \frac{\mu u}{1-\nu u}. \tag{A.2}$$

Identity return is equivalent to $\mu = 1$, $\nu = 0$, or equivalently $\mathcal{R}'(0) = 1$ and $\mathcal{R}''(0) = 0$.

Proof. Put $W = |(c' + ue') \times e|$ and choose the unit normal $N = ((c' + ue') \times e)/W$. The mixed second fundamental coefficient is Δ/W and the uu coefficient is zero. The tt numerator is the quadratic polynomial in (A.1), whose constant term vanishes because $u = 0$ is asymptotic. This gives the characteristic equation and $K = -\Delta^2/W^4 < 0$. Substitution $u = e^{A(t)}z$ gives $z' = q_2 e^A z^2$, and integration yields the displayed return. □

Without a chosen closed leaf there is also a constant term $q_0(t)$. The Riccati equation is the projectivization $u = x/y$ of

$$\begin{pmatrix} x \\ y \end{pmatrix}' = \begin{pmatrix} q_1/2 & q_0 \\ -q_2 & -q_1/2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

Its monodromy $\Phi(L)$ is the linear solution map over one period, where $\Phi(0) = \mathrm{Id}$; it lies in $\mathrm{SL}(2,\mathbb{R})$. Its projective action is $[x : y] \mapsto [\Phi(L)(x, y)]$ on $\mathbb{RP}^1$. This action is the identity exactly when $\Phi(L) = \pm\mathrm{Id}$. To obtain an affine band one must also have a finite periodic leaf, since a projective solution can pass through $u = \infty$. Near the finite leaf in the proposition this issue does not arise. For principal-normal rulings, $q_1 = -\lambda/2$, $q_2 = -D/2$, so $\mu = 1$ automatically and $\nu = -\rho(0)I_0$.

Theorem A.2 (Ruled localization as transverse transport)**.** *Under the preceding hypotheses, a sufficiently thin one-sided band $0 < u < b$ admits a nonzero compactly supported bending exactly when the return is the identity. In the identity case, the supported kernel on any saturated subband is parametrized by one arbitrary compactly supported smooth function of its leaf parameter.*

Proof. Reconstruct the field from one scalar. The uu equation and support imply $\langle e, y\rangle = 0$. For $p = \langle X_t, y\rangle$, the mixed equation becomes $\langle e', y\rangle = p_u/2$. Thus

$$\langle e, y\rangle = 0, \qquad \langle X_t, y\rangle = p, \qquad \langle e', y\rangle = p_u/2 \tag{A.3}$$

uniquely determines y, since $\Delta \neq 0$. The remaining equation is

$$p_t + qp_u = \left(q_u + \tfrac{1}{2}(\log|\Delta|)'\right) p.$$

For an explicit verification, put $a = |e'| > 0$, $f = e'/a$, and $n = e \times f$, and write $c' = \alpha e + \beta f + \gamma n$ and $f' = -ae + \omega n$. Then $\Delta = a\gamma$, and the right-hand coefficient is $(\beta\omega + \gamma' + ua\omega)/\gamma$, which agrees with the formula. Setting $h = p/\sqrt{|\Delta|}$ gives

$$h_t + qh_u = q_u h, \qquad [\partial_t + q\partial_u, h\partial_u] = 0. \tag{A.4}$$

Identify the return condition. Let $u = \phi_t(v)$ be the characteristic flow with $\phi_0(v) = v$. Then

$$h(t, \phi_t(v)) = \partial_v\phi_t(v)h(0, v), \qquad h(0, \mathcal{R}(v)) = \mathcal{R}'(v)h(0, v).$$

Thus the transported object is the transverse vector field $h\partial_u$. For identity return $v = \phi_t^{-1}(u)$ is a single-valued leaf coordinate, and any compactly supported initial function on a complete-leaf interval descends smoothly. Reconstruction gives every supported bending on that saturated subband.

If return is not the identity, shrink b to remove all fixed points in $(0, b)$: the displayed projective return has at most one other fixed point besides zero. A characteristic carrying nonzero h remains nonzero and must stay in a compact subset of the band. Its successive returns are a monotone sequence in a compact interval, and their limit would be an interior fixed point. This contradiction forces $h = 0$ and hence $y = 0$. □

Three distinct nearby closed nonruling leaves crossing a common ruling once force a general projective return to be the identity. In the principal-normal class the central leaf and one additional leaf suffice. These finite tests depend on the projective form and do not apply to arbitrary smooth surface foliations.

Appendix B. Verification of the corrugated seed

We prove the package in Lemma 5.2. The estimates serve only to give uniform visibility margins. The root and action calculations provide closure, while visibility will also imply embedding of the partner curve.

B.1. **Visibility implies embedding of a positive tangent partner.**

Lemma B.1 (Visible positive tangent partners)**.** *Let $\beta, \delta : \mathbb{R}/T\mathbb{Z} \to \mathbb{R}^2$, $T > 0$, be smooth, with β a regular positively oriented Jordan curve and $\delta' = m\beta'$, $m > 0$. If $|\delta| > R$ and $\beta' \cdot V_R(\delta) > 0$, then δ is a positively oriented Jordan curve enclosing the disk $|z| \le R$.*

Proof. The unit tangent $U = \beta'/|\beta'|$ has degree one by the turning-number theorem. Its positive scalar product with $V_R(\delta)$ makes those circle-valued maps homotopic, so $V_R(\delta)$ has degree one. Lift the parameter to $\mathbb{R}$ and write $\delta = \rho e^{i\alpha}$, with $\rho = |\delta|$ and α a continuous real angle. An angle for the visibility vector is $\varphi = \alpha + \arccos(R/\rho)$, whose second summand is periodic. Thus δ winds once around zero. Moreover

$$\varphi' = \frac{\delta' \cdot V_R(\delta)}{\sqrt{\rho^2 - R^2}} > 0. \tag{B.1}$$

The angle increases strictly by 2π in one period. Equal positions at distinct parameters would give equal angles modulo 2π, a contradiction. Hence δ is embedded and positively oriented. The disk is connected, contains zero, and misses the curve, so it lies in the bounded component. □

B.2. **Root, first curve, covariance, and action.**

For fixed N, consider the probability density proportional to $(1 + \epsilon \sin x)^2 e^{-k \cos x}$. Equation (5.4) is $\mathbb{E}_k[\cos x] = -1/2$, and

$$\frac{d}{dk}\mathbb{E}_k[\cos x] = -\operatorname{Var}_k(\cos x) < 0.$$

Concentration at the extrema of $\cos x$ gives limits 1 as $k \to -\infty$ and -1 as $k \to \infty$. Thus the root exists uniquely.

The radius of β_N is positive. If two positions agree, their radii give equal $\sin Nt$ values, after which their arguments give equal parameters modulo 2π. Thus β_N is embedded. With $c = \cos Nt$ and $s = \sin Nt$,

$$\beta_N' = e^{i(t+2\epsilon s)}\big(c + i(1 + \epsilon s)(1 + 2c)\big). \tag{B.2}$$

The real and imaginary components cannot vanish together, proving regularity. Its angle lift increases by 2π over the period, proving positive orientation and enclosure of zero.

The derivative $\delta_N' = m_N \beta_N'$ is covariant because m_N is cell-periodic. The chosen initial value gives $\delta_N(T_N) = \xi_N \delta_N(0)$, hence covariance for the whole curve and closure after N cells. Finally

$$\det(\beta_N, \delta_N') = m_N(1 + \epsilon \sin Nt)^2(1 + 2\cos Nt),$$

so the scalar root cancels the mixed action on each cell and on the full period.

B.3. **Position and tangent bounds.** Put

$$L(c) = \sqrt{c^2 + (1+2c)^2}, \qquad U_0 = e^{it}\frac{c + i(1+2c)}{L(c)}, \qquad U_N = \frac{\beta_N'}{|\beta_N'|}.$$

For $N \geq 10$ we claim

$$|\beta_N - e^{it}| \leq 3/N, \qquad |U_N - U_0| < 50/N, \qquad |\delta_N - (i/2)e^{it}| < 60/N. \tag{B.3}$$

The first uses $|e^{iu} - 1| \leq |u|$. Since $1/\sqrt{5} \leq L(c) \leq \sqrt{10}$, the unnormalized tangent in (B.2) differs from $e^{it}(c + i(1+2c))$ by less than $10/N$. The bound

$$\left|\frac{z}{|z|} - \frac{w}{|w|}\right| \leq \frac{2|z-w|}{|w|}$$

therefore gives the tangent estimate.

To control the partner, use the normalized weighted average

$$\langle f\rangle_N = \frac{1}{2\pi}\int_0^{2\pi} Z_N^{-1} e^{-k_N \cos x} f(x)\, dx.$$

Odd functions of $\sin x$ have zero average. With $y = 1 + 2\cos x$, the root identity gives

$$\langle y\rangle_N = -\epsilon^2 \langle \sin^2 x\, y\rangle_N, \quad |\langle y\rangle_N| \leq 3\epsilon^2, \quad \left|\langle \cos x + iy\rangle_N + \tfrac{1}{2}\right| \leq \tfrac{9}{2}\epsilon^2.$$

The one-cell average

$$J_N = \left\langle e^{i\epsilon(x + 2\sin x)}\big(\cos x + i(1 + \epsilon \sin x)y\big)\right\rangle_N$$

satisfies $|J_N + 1/2| < 25\epsilon$: use $0 \leq x + 2\sin x \leq 2\pi$ and the preceding average bounds. For $\alpha = 2\pi\epsilon$, write $Q(\alpha) = \alpha/(e^{i\alpha} - 1)$. For $\epsilon \leq 1/10$, $|Q| < 11/10$ and $|Q + i| < 4\epsilon$. Since $\delta_N(0) = Q(\alpha)J_N$, it follows that $|\delta_N(0) - i/2| < 30\epsilon$. Over one cell,

$$|\delta_N(t) - \delta_N(0)| \leq 2\pi\epsilon(\sqrt{10} + 3\epsilon) < 22\epsilon, \qquad |(i/2)e^{it} - i/2| < 4\epsilon.$$

Their sum is less than 60ϵ. Covariance extends this estimate to the full period, completing (B.3).

B.4. **The two visibility margins.** At $\delta_0 = (i/2)e^{it}$,

$$V_{1/4}(\delta_0) = e^{it}(-\sqrt{3}/2 + i/2).$$

Thus

$$U_0 \cdot V_{1/4}(\delta_0) = \frac{1 + (2 - \sqrt{3})c}{2L(c)} \geq \frac{\sqrt{3} - 1}{2\sqrt{10}} > \frac{1}{10}. \tag{B.4}$$

For the reflected reversed pair, the identity $A_* J = -JA_*$ reduces visibility to positivity of

$$U_N \cdot \frac{-R_-\beta_N + \sqrt{|\beta_N|^2 - R_-^2}\, J\beta_N}{|\beta_N|^2}.$$

At $\beta_0 = e^{it}$ and $R_- = 4/5$ the limiting expression is

$$\frac{-4c + 3(1+2c)}{5L(c)} = \frac{3 + 2c}{5L(c)} \geq \frac{1}{5\sqrt{10}} > \frac{1}{16}. \tag{B.5}$$

The differential norm of either direction map $z \mapsto (\pm Rz + \sqrt{|z|^2 - R^2}Jz)/|z|^2$ is $1/\sqrt{|z|^2 - R^2}$. For $N \geq 10^4$, the comparison segments in (B.3) have radii at least 0.494 for the outer test and 0.9997 for the inner test. A Lipschitz bound of 3 therefore suffices. The respective errors are at most $50/N + 3(60/N) = 230/N$ and $50/N + 3(3/N) = 59/N$. Both are smaller than their displayed margins, and both radius inequalities hold. Since the velocities in each pair are positive multiples, both required tangent inequalities follow. The positive-partner lemma proves embedding and enclosure for δ_N.

B.5. **The spherical curvature is nonconstant.** The gnomonic lift is embedded and stays in the northern hemisphere. With the sign convention (5.8), its geodesic curvature has the sign of $\det(\beta_N', \beta_N'')$. Writing $r = 1 + \epsilon \sin Nt$, $c = \cos Nt$, and $s = \sin Nt$, differentiation gives

$$\det(\beta_N', \beta_N'') = Nrs + 2c^2(1 + 2c) + r^2(1 + 2c)^3.$$

The values at $Nt = 0$ and $Nt = \pi$ are 33 and -3. The curvature therefore takes both signs. This finishes every assertion of Lemma 5.2.

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